MILITARY SPECIFICATION

RECORDER/REPRODUCER, ACOUSTIC AN/AQH-4(V)2

This specification is approved for use by the Naval Air Systems Command Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers, requirements for the AN/AQH-4(V)2 Recorder/Reproducer, refered to herein as the equipment.

1.2 Associated Equipment. This equipment should operate with the associate equipment listed in 6.8.

2. APPLICABLE DOCUMENTS.

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Dept. (ESSD), Code 93, Lakehurst, N.J. 08733) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5835

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SPECIFICATIONS

FEDERAL

W-T-001553	Tape, Recording, Instrumentation, Magnetic Oxide Coated
W-R-175	Reel And Hub For Magnetic Recording Tape, General Specification For
W-R-175/4	Reels, Precision, Aluminum, With 3-Inch Center Hole (Detail Specification)
W-R-175/6	Reel. Precision, Glass Flange With Aluminum Hub, 3-Inch Center Hole (Detailed Specification)

MILITARY

MIL-E-5400	Electronic Equipment, Aerospace, General specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic and Associated Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical and Aeronautical upport Equipment Including Ground Support Equipment
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-C-26482	Connector, Electrical, (Circular, Miniature Quick Disconnect, Environment Resisting) Receptacles and Plugs, General Specification for
MIL-M-38510	Microcircuits, General Specification for
MIL-C-81511	Connector, Electrical, Circular, High Density Quick Disconnect, Environment Resisting, and Accessories, General Specification for

STANDARDS

FEDERAL

FED-STD-595	Color

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MIL-STD-415	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-454	Standard General Requirement for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Verification/Demonstration/Evaluation
MIL-STD-701	Lists of Standard Semiconductor Devices
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-883	Test Methods and Procedures for Microelectronics
DOD-STD-1460	Soldering of Electrical Connections and Printed Wiring Assemblies Procedure for
MIL-STD-1610	Magnetic Tape Recorder/Reproducer Formats for ASW Sensor Data
MIL-STD-2068	Reliability Development Tests
MIL-STD-2074	Failure Classification for Reliability Testing
MIL-STD-2084	General Requirements for Maintainability of Avionic and Electronic Systems and Equipment

2.1.2 Other government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein.

NAVAL AIR SYSTEMS COMMAND

WR-101 PART I	Electromagnetic Control Requirements for Advanced ASW Avionics Systems

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

INTER-RANGE INSTRUMENTATION GROUP

IRIG	104	Standard Time Code
IRIG	106	Telemetry Standards
IRIG	118	Test Methods For Telemetry Systems

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.2 and 6.4)

3.2 Parts, materials, processes, and standards. In the selection of parts, materials, processes and standards, the requirements of MIL-E-5400 shall govern. The following items require particular attention and are subject to the provisions indicated below:

a. The use of spot welding shall receive the procuring activity's prior approval, unless previously used on AQH-4 design.

b. Nonrepairable subassemblies shall be used as practicable and approved in accordance with MIL-E-5400.

c. Resistors shall be MIL-spec type and capacitors shall be established reliability type, unless otherwise approved by the procuring activity.

d. MIL-M-38510 microcircuits shall be used as first order of precedence. Any nonstandard microcircuit shall be available from two or more sources and shall be screened and conform to MIL-STD-883, Methods 5004 and 5005 Class B. Precap inspection of all microcircuits is required. Devices selected shall be soldered or welded into their circuits.

e. The use of the following is prohibited:

(1) Magnesium.

(2) Nonhermetically sealed, wet tantalum capacitors and aluminum oxide capacitors.

(3) Metallic oxide rectifiers.

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3.2.1 Nonstandard parts and materials approval. Approval for the use of nonstandard parts and materials, including diodes, transistors, microelectronics and nonrepairable subassemblies, shall be obtained in accordance with MIL-E-5400. Nonstandard microcircuits shall have a minimum of two sources of supply. In exceptional cases, when it is essential to use a nonstandard microcircuit having but a single source, the contractor shall obtain procuring activity approval for such use.

3.2.2 Nonrepairable assemblies. All nonrepairable assemblies are considered to be nonstandard parts. A listing of the nonrepairable assemblies shall be prepared and made available to the procuring activity (see 6.2.2).

3.2.3 Electron tubes. Electron tubes shall not be used in this equipment unless the contractor shows that the function cannot practicably be accomplished by any other device. If tubes are used, the equipment shall meet the electron tube requirements of MIL-STD-454, Requirement 29.

3.2.4 Semiconductor devices. The equipment shall meet the semiconductor devices requirements of MIL-STD-454, Requirement 30. Transistors and diodes shall be JAN TX type, where available, or JAN type; and chosen and applied in accordance with the requirements of MIL-S-19500. Whenever possible, semiconductor devices shall be those listed in MIL-STD-701. Non-JAN type semiconductor devices are considered to be nonstandard parts and shall be approved by the procuring activity. (see 6.2.2).

3.2.5 Module interchangeability. Modules of like type (see 6.5) shall be interchangeable and identical with all other modules of that type for all equipments produced in accordance with this specification.

3.3 Design and construction. The equipment shall conform with all the applicable requirements of MIL-E-5400 for Class 1A equipment for design, construction, and workmanship, except as otherwise specified herein.

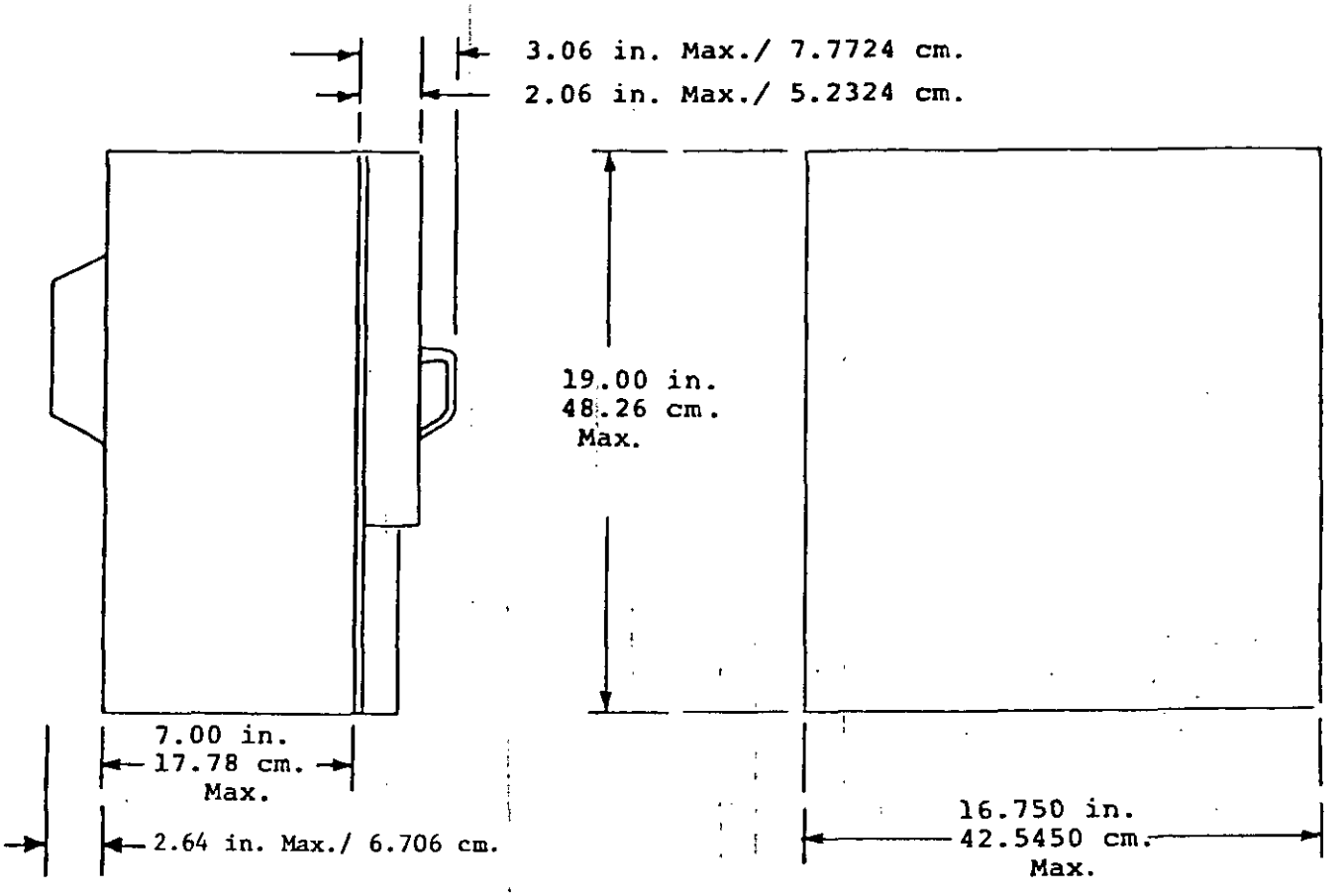
3.3.1 Total weight. The total weight of the equipment, excluding tape and reels, shall be a minimum consistent with good design, and shall be not greater than 76 pounds.

3.3.1.1 Form factor. The AN/AQH-4(V)2 Recorder/Reproducer shall be not greater than the following dimensions (except the handles may project one inch):

Width	Height	Depth
16.75 inches	19 inches	11.7 inches

3.3.1.2 Interchangeability. The AN/AQH-4(V)2 Recorder/Reproducer shall be mechanically interchangeable, except for connectors, with the AN/AQH-4(V) Recorder/Reproducer (see FIGURE 1).

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Electrical Connectors:	Location of connectors not critical except that they be accessible from rear of unit.
Primary Power Inlet:	J1 Receptacle to mate with MS3126E14-12S
Remote Control:	J2 Receptacle to mate with MS3126E18-32S
Data Input:	J3 Receptacle to mate with M81511/6EE01S1
Data Output:	J4 Receptacle to mate with M81511/06EE01S3

FIGURE 1 Envelope and Connectors, AN/AQH-4(V)2.

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3.3.1.3 Thermal design. The thermal design of the equipment shall be such that, with an ambient air temperature of 54.5°C, no solid state component or capacitor case temperature shall exceed 86.7°C, except for the bias oscillator transistors, which shall not exceed 96.1°C. Semiconductor devices and integrated circuit junction temperatures shall be not greater than 110°C.

3.3.2 Reliability

3.3.2.1 Reliability program. The contractor shall establish and conduct a reliability assurance program as specified in 3.3.2.2 through 3.3.2.6.1.

3.3.2.2 Operational stability. The equipment shall operate continuously or intermittently for a period of at least 200 hours without readjustment of any controls which are inaccessible to the operator during normal use. Such readjustment includes tape path realignment. Normal tape resupply, tape path cleaning or degaussing is not to be interpreted as a readjustment.

3.3.2.3 Operating life. The equipment shall have a total operating life of 10,000 hours with minimal service and replacement of parts.

3.3.2.4 Reliability in mean time between failure (MTBF). The mean (operating) time between failures for the equipment, including BITE, shall meet or exceed 500 hours.

3.3.2.4.1 Reliability guarantee. All malfunctions occuring as a result of the testing identified herein shall be failure diagnosed (repair analysis) to identify the failed piece parts, and reported in writing to the procuring activity (see 6 2 2). This function shall be conducted by the contractor at his expense at his facility. If a design change is required as a result of failure during testing, then corrective action shall include incorporation of the change(s) in all preproduction and production units.

3.3.2.5 Time totalizing meters. The equipment shall contain two time totalizing meters in accordance with MIL-M-7793. The meters, (10,000 hours each), shall be included in the equipment; one meter shall monitor total system operating time, and the other shall monitor total tape travel time. Each meter shall be clearly identified regarding its respective function, and shall be visible when the unit is installed in the normal configuration.

3.3.2.6 Failure reporting analysis and corrective actions. The contractor shall implement a closed loop failure reporting analysis and corrective action program. The contractor shall provide for visibility and traceability of all reported failures, beginning with those occurring during final assembly checkout and ending with acceptance testing.

3.3.2.6.1 Failure. A failure shall be classified as the inability of the equipment to meet the requirements of the equipment specification, including excessive wear and misalignment of tape path components. All unscheduled maintenance or adjustment shall constitute a failure.

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3.3.3 Cabling and connectors. The equipment shall provide for the use of cables and connectors as follows:

3.3.3.1 Interconnection cabling. The equipment shall operate as required using external wiring in accordance with the applicable requirements of WR-101, Part I. The external wiring shall be unshielded, except that a minimum number of individual wires may be shielded when demonstrated as necessary to meet interference control requirements. External cables, and that portion of the connectors attached to the cables, shall not be supplied as part of the equipment. Pin assignments on each external connector shall be in accordance with FIGURES 2 thru 5 herein.

3.3.3.2 Electrical connectors. Electrical connectors shall be in accordance with MIL-C-26482 and MIL-C-81511, as specified in FIGURE 1.

3.3.3.3 Ground connections. Ground circuit design shall be in accordance with the following:

a. A separate power return shall be brought out to an external connector pin and not be grounded within the equipment

b. Safety ground shall be brought out to an external connector pin.

c. Signal ground shall be brought out to an external connector pin.

d. Use of common grounding wires within the equipment for signal grounds and DC power grounds shall be avoided.

3.3.4 Control panels. All rack or console mounted control panels shall perform to the applicable requirements of MIL-C-6781, except that the panel shall be gray in accordance with FED-STD-595 (color No. 36231) and shall have amber lights.

3.3.4.1 Light sources. All lamps shall be 28 V lamps. The lamps shall be supplied with a fixed voltage of 18 VDC. All lamps shall be removable. Removal of lamps shall not require disassembly of the equipment. Indicator lights shall be tested with 28 VDC.

3.3.5 Interchangeability of reordered equipment. For reordered equipment, interchangeability shall exist between units and all replaceable assemblies, subassemblies, and parts of a designated model of any previously manufactured equipment, supplied or designated by the procuring activity.

3.3.6 Interference control. The equipment shall conform with all of the requirements of MIL-STD-461, Class 1C.

3.3.7 Maintainability

3.3.7.1 Maintainability program. The contractor shall establish and conduct a maintainability program in accordance with MIL-STD-2084.

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J1

(MS3122E-14-12P) OR

(MS3112E-14-12P)

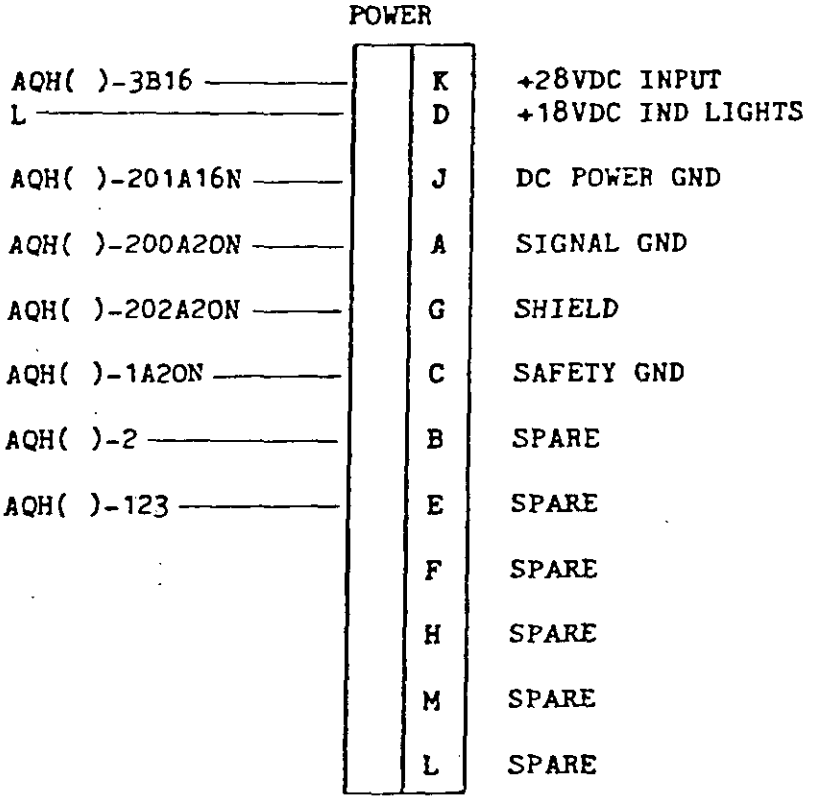


FIGURE 2 AN/AQH-4(V)2 Electrical connector, power.

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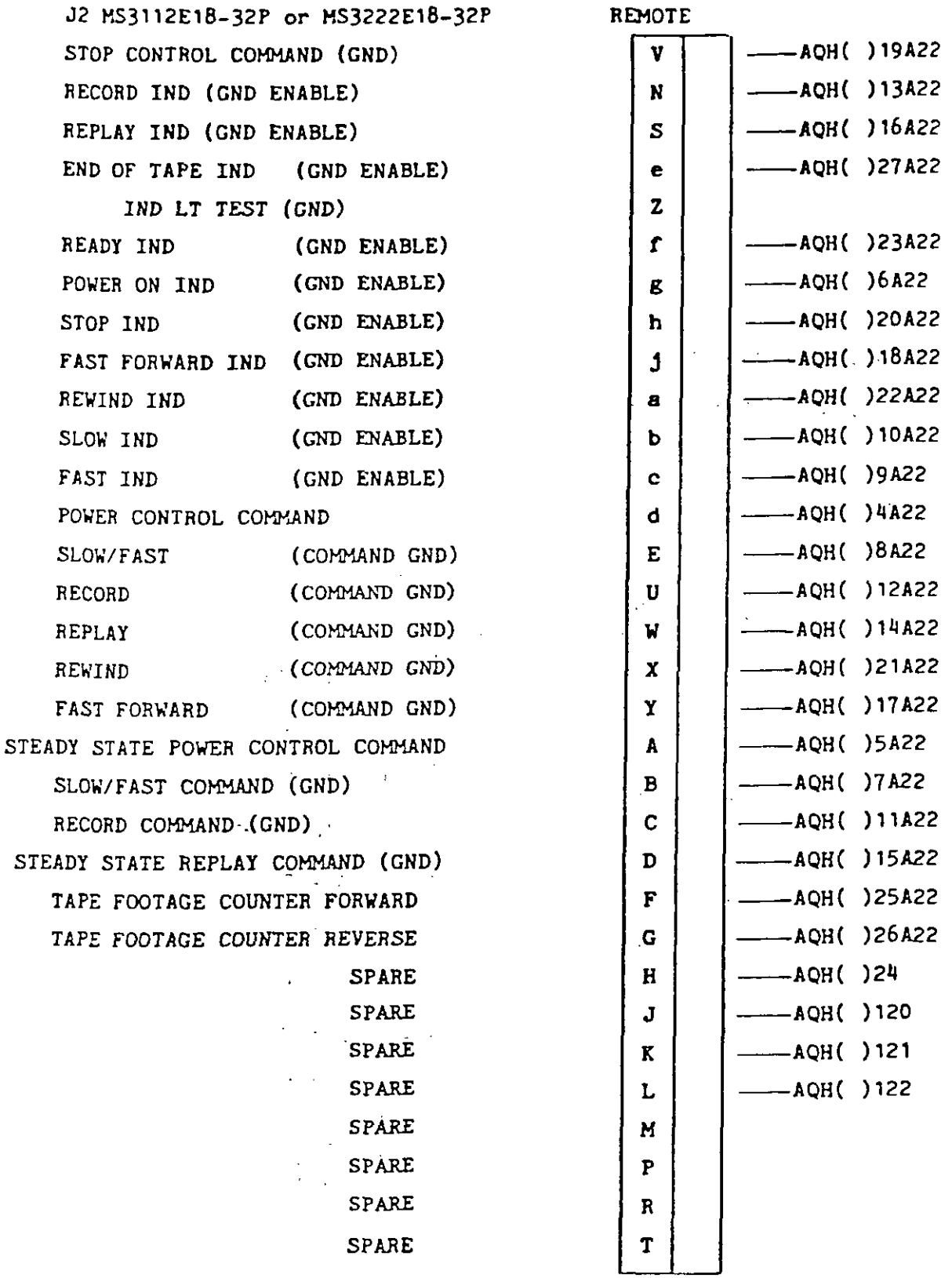
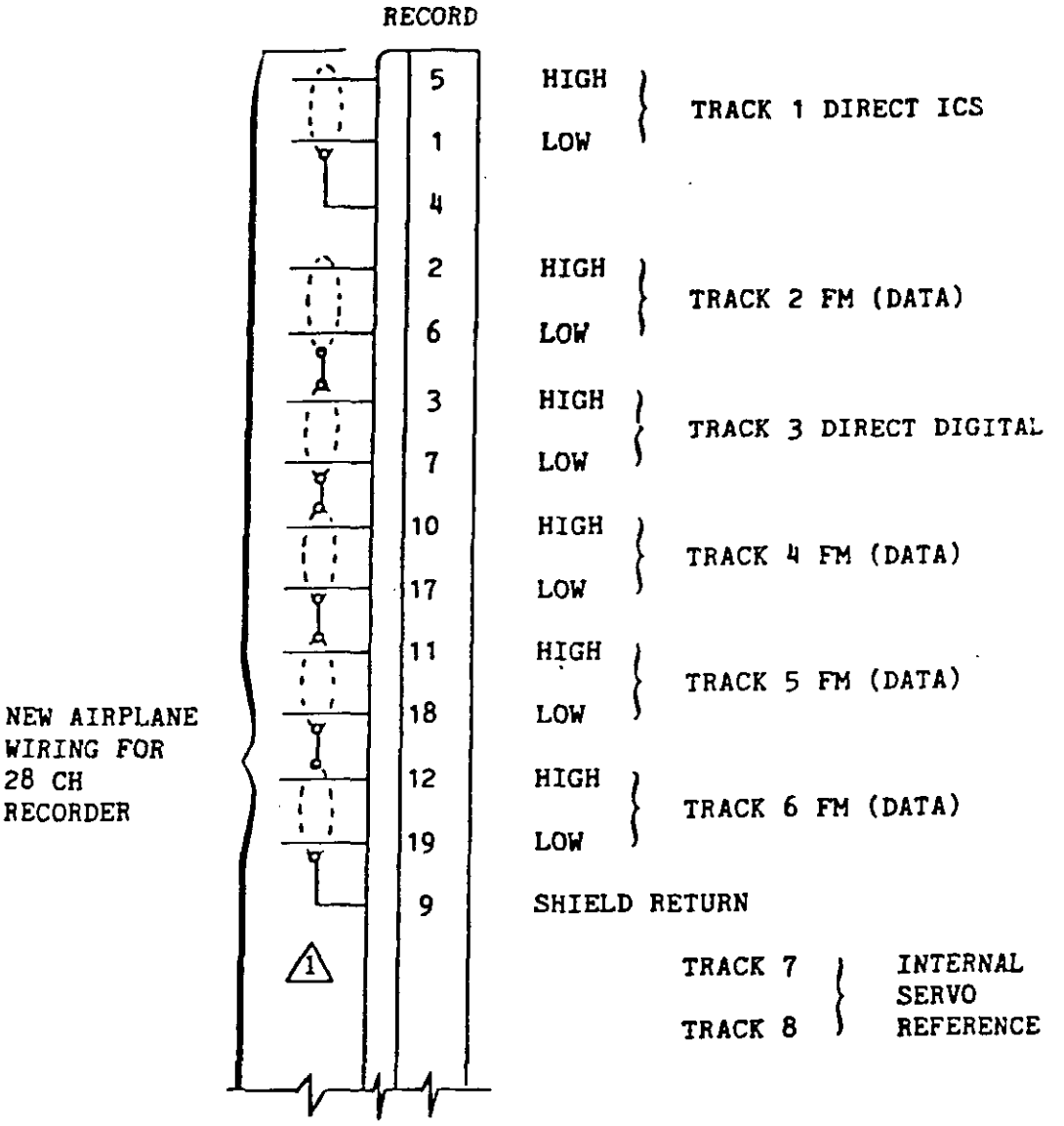


FIGURE 3 AN/AQH-4(V)2 Electrical connector, remote.

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J3

(M81511/01EE01P1)



1 FIGURE NO. 4C EXTERNAL SHIELDED WIRING MAY BE DELETED FROM AIRCRAFT INSTALLATIONS BASED ON EMI TEST RESULTS

FIGURE 4 AN/AQH-4(V)2 Electrical connector, record.

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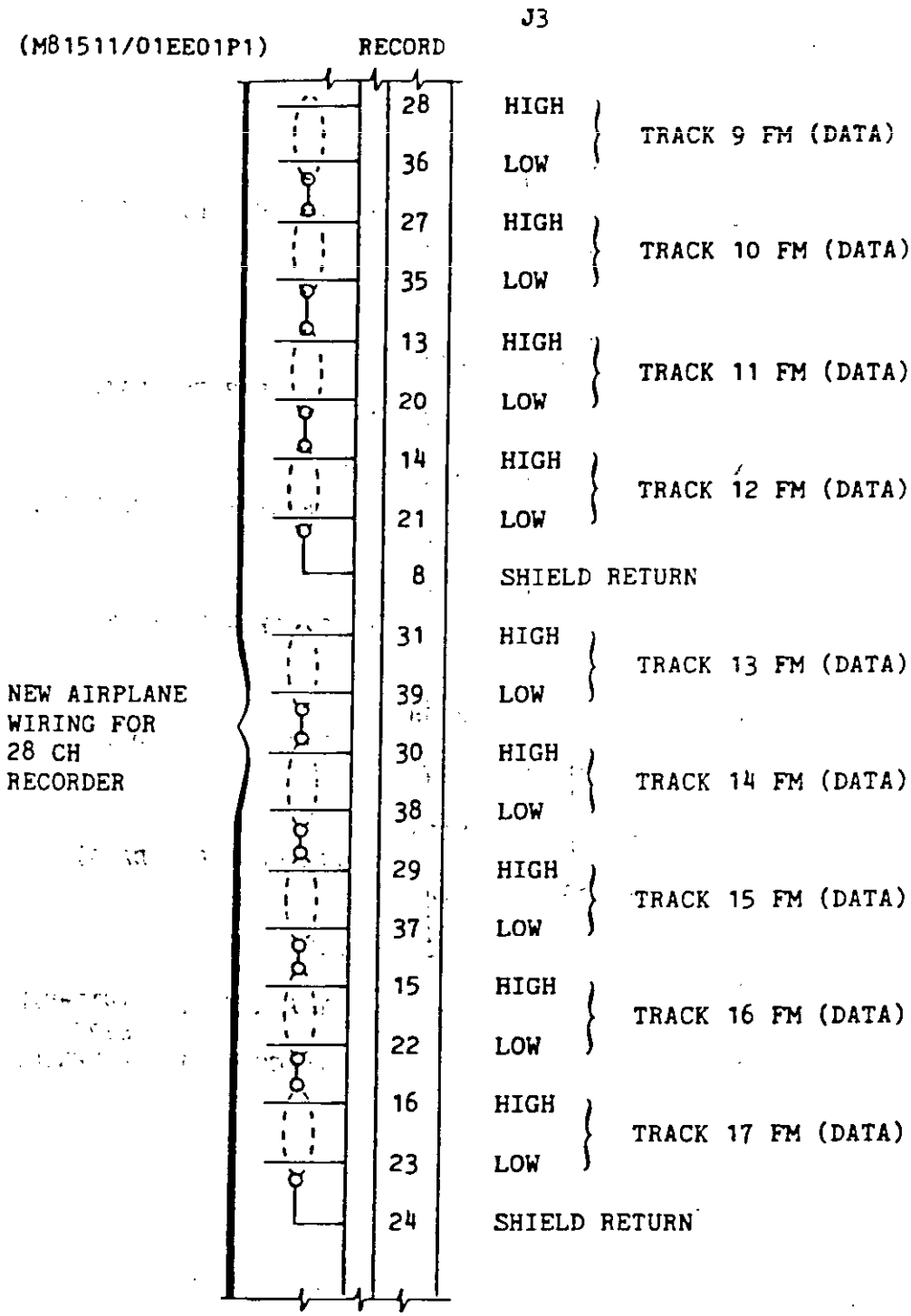


FIGURE 4 AN/AQH-4(V)2 Electrical connector, record. (Continued)

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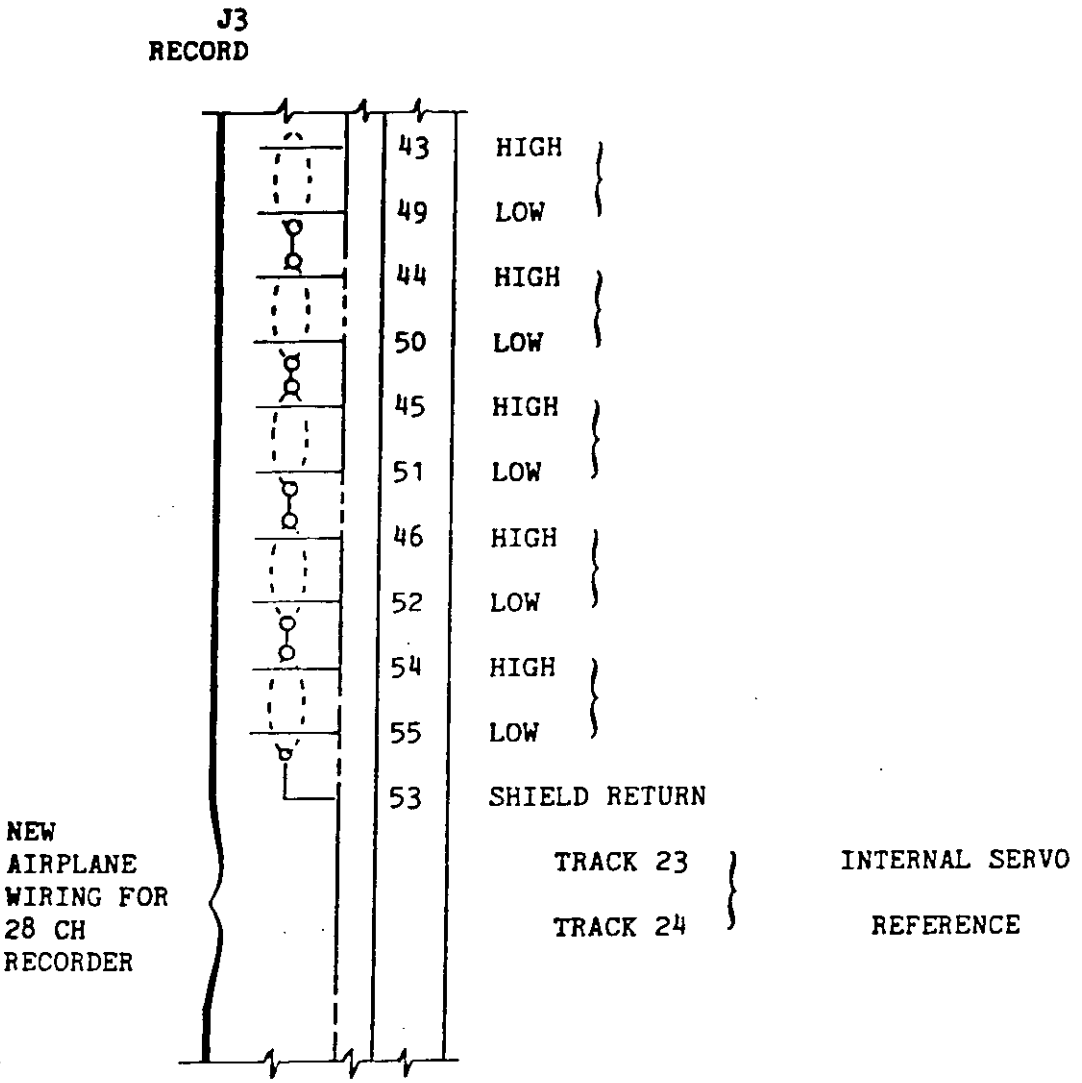


FIGURE 4 AN/AQH-4(V)2 Electrical connector, record. (Continued)

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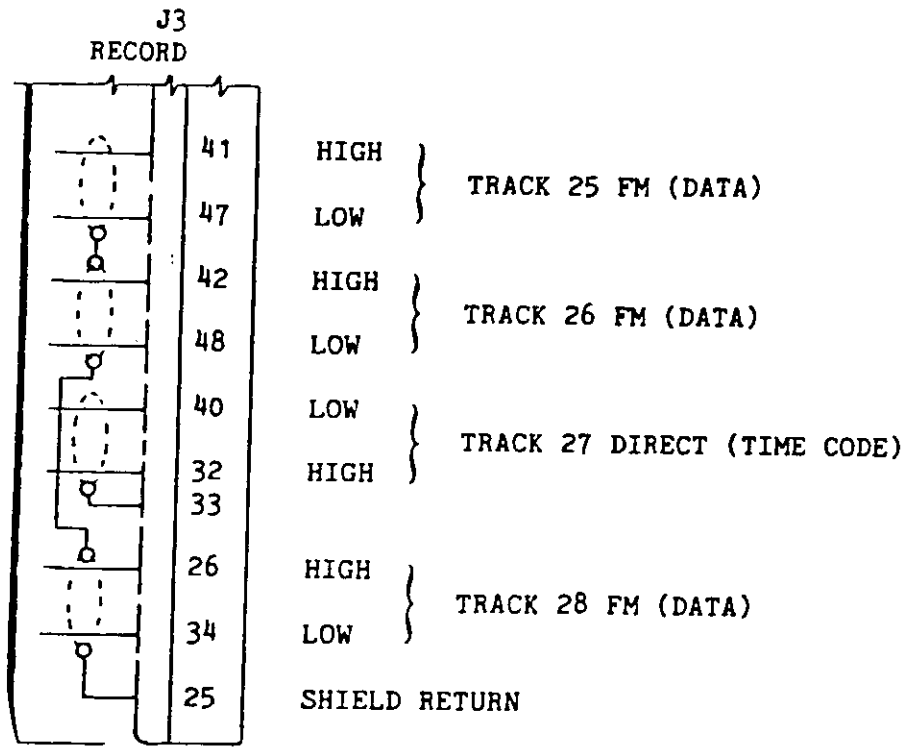


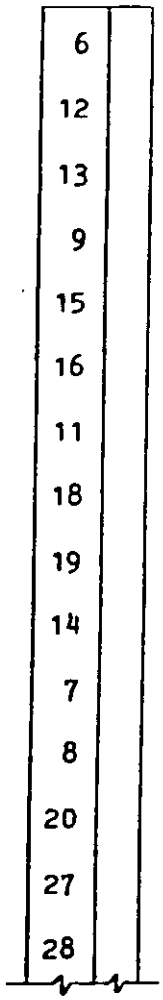
FIGURE 4 AN/AQH-4(V)2 Electrical connector, record. (Continued)

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J4

(M81511/01EE01P3)

REPLAY



FOR GROWTH
THIS SECTION
INTENTIONALLY
LEFT BLANK

FIGURE 5 AN/AQH-4(V)2 Electrical connector, replay.

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J4

(M81511/01EE01P3)

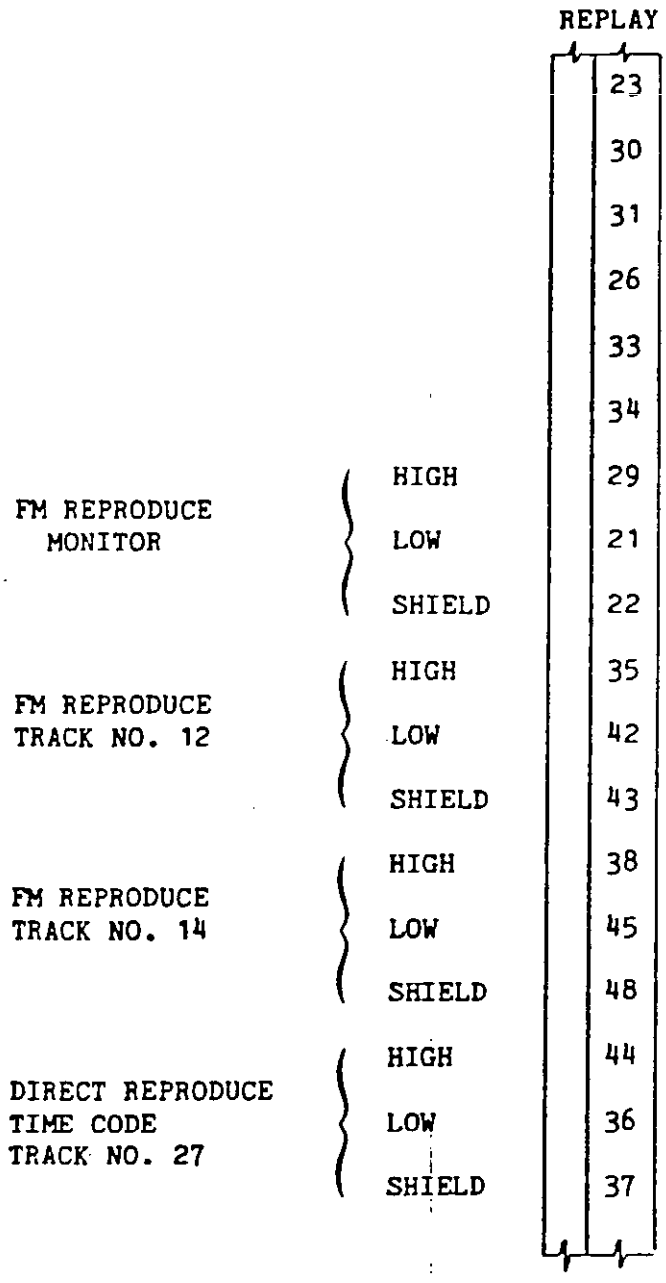


FIGURE 5 AN/AQH-4(V)2 Electrical connector, replay. (Continued)

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J4

(M81511/01EE01P3)

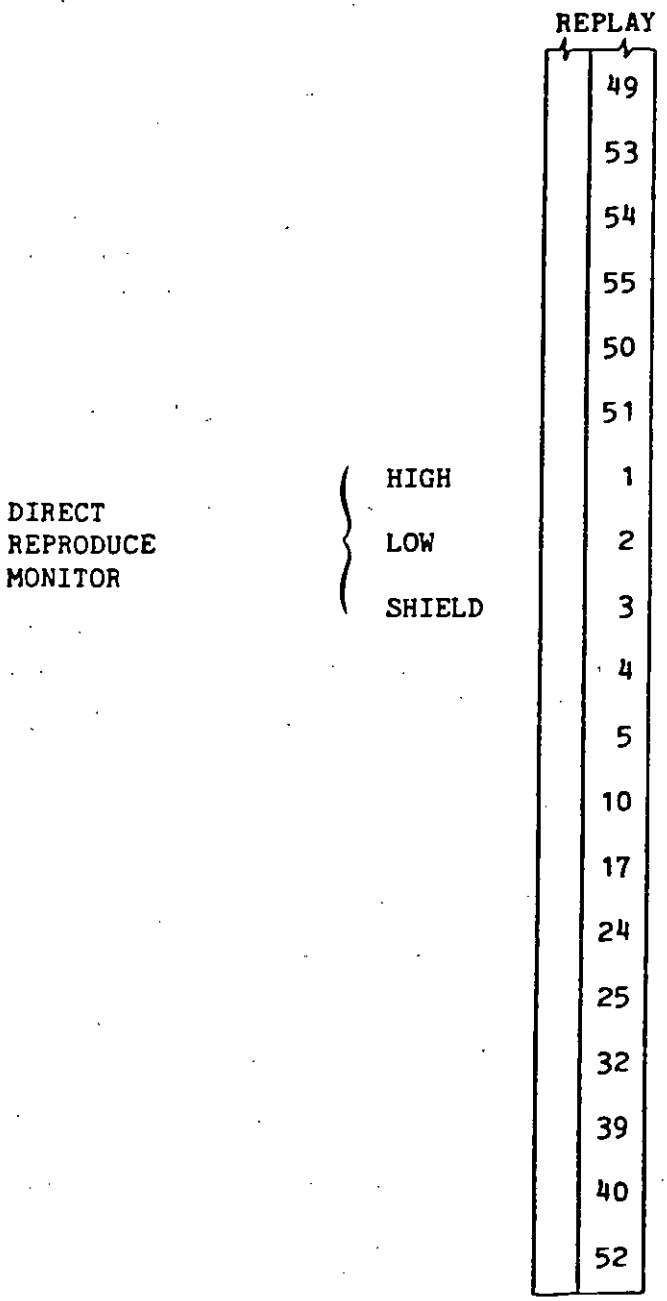


FIGURE 5 AN/AQH-4(V)2 Electrical connector, replay. (Continued)

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3.3.7.1.1 Packaging requirements. The equipment shall be packaged in modular form so that, to the maximum extent practical, all subassemblies shall be the plug-in type and shall be easily removable. If a special tool is required for removal, it shall be provided and attached to the equipment. All organizational maintenance level replacement items shall be retained by captive hardware.

3.3.7.1.1.1 Module accessibility. Module arrangement shall be such that access to each module does not require the removal of adjacent modules or parts other than access panels.

3.3.7.1.2 Keying. Modules and connectors shall be keyed to prevent the insertion of a connector or module into an improper location within or on the equipment.

3.3.7.1.3. Factory adjustments. Factory adjustments, such as those necessary to match head assemblies to other maintenance modules, shall be located on the chassis, and not on the affected maintenance module. These provisions shall negate the need for organizational level readjustment each time the maintenance modules are replaced or interchanged.

3.3.7.1.4 Access panels. Access panels shall be secured by captive quick disconnect fasteners.

3.3.7.1.5 Service access. Devices requiring periodic removal for inspection or servicing shall be secured by captive quick disconnect fasteners.

3.3.7.1.6 Equipment readiness measurement. Suitable features shall be incorporated in the equipment to provide the operator seated at his station, while airborne or on the ground, with a GO/NO-GO indication of equipment readiness. Detailed GO/NO-GO data is specified in 3.3.7.1.7 and 3.4 2 4.1.

3.3.7.1.6.1 Mode of operation. The operation of the equipment readiness test feature shall be accomplished manually.

3.3.7.1.7 Fault isolation. In conjunction with, but not necessarily as part of, the performance/readiness testing features, there shall be features incorporated which will permit fault isolation to the module level. These features shall include the necessary switching arrangements with a GO/NO-GO readout device that will clearly indicate whether or not the function selected is within minimum performance standards specified herein. This capability shall be provided by the use of Built-In-Test-Equipment (BITE) to interrogate every signal channel for both amplitude and frequency, including the servo reference, time code, ICS and digital channel 1/. A front panel switch and associated GO/NO-GO indicator shall allow the operator to select any record/monitor/reproduce channel for test, thus automatically introducing a test signal into the channel and providing a visual indication of functional capability. Comparative checks of channels shall allow gross estimates of fault isolation of the record channels, monitor reproduce

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functions, or record/reproduce heads. When the BITE/normal selector switch is in the normal position, a GO indication on the GO/NO-GO indicator shall demonstrate availability of a test signal for BITE use; when the BITE signal is applied to a functional channel, the GO/NO-GO indicator shall provide a NO-GO indication for the brief interval during which tape is reeled from the record to reproduce heads, and then, provide the GO indication when the proper reproduce signal is recognized by the BITE. Upon activation of the BITE test, all record channels shall simultaneously commence recording of the BITE test frequency.

3.3.7.1.8 Test points. All test measurements shall be made through test points. No test connectors shall be used. Test points shall accommodate standard government approved test equipment, i.e., multimeter or oscilloscope probes. Adapters are permitted, provided they are included as an integral part of the equipment. The ground reference test points associated with any voltage test points shall be clearly marked with respect to their function and association with those specific voltage test points. All test points shall be in accordance with MIL-STD-415, except as modified herein.

3.3.7.1.8.1 Record/reproduce module test points. Input and output signal test points for each record and reproduce amplifier shall be front accessible without requiring removal of access panels, the equipment from its installation, or any surrounding equipment or panels. These test points shall be in accordance with 3.3.7.1.8.1.1 through 3.3.7.1.8.1.5.

3.3.7.1.8.1.1 FM record test points. As a minimum, there shall be test points for each channel of an FM record module to allow inspection of signals at the following locations in the signal path:

a. Channel Input.

b. VCO Output (Voltage Controlled Oscillator).

c. Channel Output.

d. Ground Reference.

3.3.7.1.8.1.2 Direct record test points. As a minimum, there shall be test points for each channel of a direct record module to allow inspection of signals at the following locations in the signal path:

a. Channel Input.

1/ Growth potential only.

b. Channel Output.

c. Ground Reference.

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3.3.7.1.8.1.3 FM reproduce test points. As a minimum, there shall be test points for each channel of FM reproduce electronics to allow inspection of signals at the following locations in the signal path:

a. Channel Input.

b. Limiter Output.

c. Discriminator Output.

d. Channel Output.

e. Ground Reference.

3.3.7.1.8.1.4 Direct reproduce test points. As a minimum, there shall be test points for the direct reproduce electronics to allow inspection of signals at the following locations in the signal path:

a. Signal Input.

b. Signal Output.

c. Ground Reference.

3.3.7.1.8.1.5 Digital record test points 1/. As a minimum, there shall be test points for each channel of the digital record module to allow inspection of signals at the following locations in the signal path:

a. Channel Input.

b. Buffer Output.

c. Channel Output.

d. Ground Reference.

3.3.7.1.8.2 Power supply test points. Power supply voltages shall be available for measurement at test points. All power supply test points shall be easily accessible without removing access panels, the equipment from its installation, or any surrounding equipment or panels.

3.3.7.1.8.3 Reference frequencies test points. There shall be test points for record bias, record servo reference, phase comparator output and frequencies, as well as for their associated ground references.

1/ Growth potential only.

3.3.7.1.9 Maintainability assurance. Testing to demonstrate the achievement of the maintainability requirements specified herein shall be in accordance with 4.5

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3.3.7.2 Reference designations. Reference designations shall be employed in accordance with the requirements of MIL-E-5400, except that the existing connectors A2JA1 through A2JA14 on the main frame may retain these designations for commonality with previous equipment.

3.3.8 Nomenclature and nameplates. Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.8.1 Nomenclature for assemblies/subassemblies and modules. Assemblies/subassemblies and modules shall be marked with their proper designator, assembly/subassembly or module group, revision number, and manufacturer's code, etc. It shall not be necessary to remove components from the assembly/subassembly or module to read those markings. The CARD cage slot shall be numerically marked in accordance with the CARD designation label.

3.3.9 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

a.	Temperature	Room ambient (25°C ±5°C)
b.	Altitude	Normal Ground
c.	Vibration	None
d.	Humidity	Room ambient (40%) up to 90% Relative Humidity
e.	Input Voltage	27.5 0.5Vdc

3.3.10 Service conditions. The equipment shall operate as required under any of the environmental service conditions or combination of those conditions as specified in MIL-E-5400 for Class 1A equipment, except as modified herein.

3.3.10.1 Vibration. The equipment shall operate as required when subjected to the vibration requirements of Curve II of MIL-E-5400. The equipment shall not require external vibration isolators.

3.3.10.2 Altitude - The equipment shall operate as required when subjected to an altitude of 4,572 meters (15,000 feet).

3.3.10.3 Temperature

3.3.10.3.1 Operating temperature. The equipment, including tape qualified to W-T-001553(2), shall operate within required limits over the temperature range of -0°C minimum to +50°C maximum. The equipment, with dummy tape, shall be capable of continuous functional operation over the temperature range of -20°C ±2°C to +55°C, without component failure, while in record or reproduce modes, at any tape speed up to 30 ips.

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Note: Any failure determined to be due exclusively to tape shall be considered nonchargeable.

3.3.10.3.2 Storage temperature. The equipment, less tape, shall withstand, without damage, storage temperatures through the range of -54°C to +85°C. After being exposed to temperatures that include these limits, the equipment, with tape qualified to W-T-001553(2), shall operate within required limits over the temperature range of 0°C to +50°C.

3.3.11 Warm-up time. The time required for the equipment to warm-up prior to operation shall be kept to a minumum and shall be not greater than two minutes under standard conditions, five minutes at +10°C and 10 minutes at -10°C.

3.3.12 Primary input power requirements. The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources with characteristics and limits as in MIL-STD-704. The equipment shall operate within the bounds of limits two and three, FIGURE 9, of MIL-STD-704, Change 3. Degraded operation may occur when the input voltage exceeds the above limits, but remains within limits one and four of MIL-STD-704. No damage shall be suffered by the equipment under transient conditions bounded by limits one and four, and normal operation shall be resumed automatically upon return of the input voltage to levels within limits two and three. In the event of power interruption, the recorder shall hold the mode it is in for 100 milliseconds. If the power loss is in excess of 100 milliseconds, the recorder shall go to the stop-mode and remain there until another mode is initiated by the operator after power has returned.

3.3.12.1 Operating power. The power required shall be not greater than the following amounts:

DC Power, 28V, Category "B", 300 watts

3.3.12.2 Lighting power. The power required shall be not greater than the following amounts:

DC Power, 18V, Category "B" for annunciator and advisory lamps

3.3.12.3 Circuit protection. The equipment shall provide for short circuit and overvoltage protection for internal circuitry without the use of fuses.

3.3.13 Cooling. The equipment shall be designed to meet the cooling requirements of MIL-E-5400, except that DC power shall be used. The equipment shall be subjected to thermal performance tests to determine the effectiveness of the cooling method in reducing component temperatures.

3.3.13.1 Fans. Air inlets and outlets shall be shown on the outline and mounting drawing and shall be fully dimensioned.

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3.3.13.1.2 Acoustical noise. When the equipment is operating in accordance with normal operating procedures, the fan shall not emit sound power in excess of the levels specified in Table I. The sound pressure spectrum level for any pure tone within any octave band shall not exceed the levels specified in TABLE II.

TABLE I. Acoustical noise - sound power level.

Octave Band Center Frequency, Hz	63	125	250	500	1000	2000	4000	8000
Sound Power Level, L_w db Re: 10^{-12} watt	90	84	77	76	72	69	66.5	64

TABLE II. Acoustical noise - sound pressure level.

Octave Band Center Frequency, Hz	63	125	250	500	1000	2000	4000	8000
Sound Pressure Level, L_p db Re: 0.0002 bar	80	74	67	74	66	66	59	59

3.4 Performance. Unless otherwise specified in the contract, (see 6.2.1) values shall be set forth to establish the requirements for satisfactory service conditions when using magnetic tape in accordance with W-T-001553, wound on precision metal or glass reels, and supplied to the requirements of W-R-175. Reduced performance beyond the temperature range of 0°C and 50°C shall be acceptable only if it is due to tape performance limitations. Unless otherwise specified herein, all values of voltage and frequency within the performance limits are allowable for determining full compliance with this specification.

3.4.1 Operation. The equipment shall be capable of recording on all 28 channels simultaneously. Concurrently, and without affecting the record mode, the unit shall reproduce simultaneously, on three fixed-wired FM channels, one direct channel, and one selectable, monitor reproduce channel. The monitor reproduce channel shall have the capability of being switched to any one of the reproduce tracks. Each channel shall be independant of any other channel, and there shall be minimal interaction between channels.

3.4.1.1 Phase-locked servo system. This equipment shall employ a capstan/tape phase-locked servo system for accurate speed control and flutter reduction. This includes tape phase-lock to a prerecorded tape made from other AN/AQH-4(V)2 equipment procured under this specification.

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3.4.1.2 Record electronics. The Record electronics shall utilize additive bias record techniques for all recording. Saturated recording is prohibited.

3.4.1.2.1 FM record. The FM record electronics shall use MIL-STD-1610 Wideband group II carrier frequencies, deviations and signal bandwidths.

3.4.1.2.2 Direct record. The Direct record method shall be used for servo reference frequencies, ICS, Time Code, and potential serial digital pulse data.

3.4.1.3 Reproduce electronics. The reproduce electronics shall be comprised of three hardwired FM reproduce channels, one hardwired direct reproduce channel for time code, and one switchable monitor reproduce channel.

3.4.1.3.1 FM reproduce. The FM reproduce electronics shall be provided for replay of tapes at 7-1/2 ips and 1-7/8 ips.

3.4.1.3.2 Direct reproduce. The Direct reproduce electronics shall be provided for the replay of direct recorded time code data, ICS and servo reference.

3.4.1.3.3 FM/direct monitor reproduce. The FM/direct monitor reproduce electronics shall be provided for monitoring the recording of signals on each individually selectable track. The monitor shall allow for the verification that each signal is being recorded properly.

3.4.1.4 Record/reproduce heads. The record and reproduce heads shall conform to MIL-STD-1610, Format A. These heads shall allow for the recording and reproducing of signals recorded directly within the frequency bands indicated for wideband 2 MHz machines, under Table 7 of IRIG Document 106 for tape speeds of 3 3/4 ips through 30 ips. Heads shall be certified for compliance to this specification prior to installation into the AN/AQH-4(V)2.

3.4.1.5 Sequence. The AN/AQH-4(V)2 Recorder/Reproducer shall provide for automatic sequencing of additional recorder/reproducer units. It shall also be possible to loop-sequence recorder/reproducer units by interconnection of the remote control connectors (see 3.4.2.4.3.3.3). The sequence command shall be issued upon sensing low tape, broken tape or any unprogrammed stop.

3.4.1.6 Expansion and growth. The equipment provided in accordance with this specification shall be designed in such a way to allow the ability to handle increased quantities of digital record with concurrent decreases in the quantity of FM record electronics.

3.4.2 Record/Reproducer speed requirements. The equipment shall be capable of recording and replaying at 1-7/8, 3-3/4, 7-1/2, 15, and 30 ips.

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3.4.2.1 Tape speeds. The equipment shall be capable of normal operation at 1-7/8 and 7-1/2 ips. It shall be possible to electrically select between either of the two provided speeds locally or at a remote location. The higher speed shall be enabled automatically when the transport is energized after a power shutdown.

3.4.2.1.1 Speed accuracy. Under capstan tape and tachometer phase-locked servo control, the tape speed accuracy shall be 0.2 percent. With a tape phase-lock servo, the tape speed stability shall be within ±0.002 percent in accordance with MIL-STD-1610. Speed accuracy shall be measured in accordance with IRIG 118.

3.4.2.2 Start time. With the tape properly loaded, the equipment ready, the warm-up conditions met, and with the actuation of the record function, stable tape operation shall be realized and the equipment shall perform within the limits of normal operating requirements as follows:

	Tape Speed	Start Time
a.	1-7/8 ips	3.0 seconds
b.	7-1/2 ips	5.0 seconds

3.4.2.3 Channel configuration. The required configuration of record channels is listed in TABLE III according to signal type.

3.4.2.4 Controls and indicators. The following controls, indicators, provisions to accept remote control signals, and provisions for operating remote indicators shall be provided.

3.4.2.4.1 Control module. A channel selector switch, and eight pushbutton, momentary contact switches with legends as indicated below, shall be located on the control module for control and display. The control signal in each case shall be a momentary ground potential. All nine controls shall be readily accessible without the need to open or remove any doors or access panels. The integral indicator lights in the switches shall illuminate to indicate status and selected or active modes.

a. POWER-READY - This switch shall control power to the equipment and provide two status indications through dual legends. When power has been turned on by depressing the switch, the power legend shall illuminate. Illumination of the power ready legend indicates that the equipment temperature is above the low temperature operating limit, and the end-of-tape (and tape break) sensor switch senses that tape is installed and tape tension is satisfactory for initializing operation in accordance with specified performance.

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TABLE III. Channel configuration.

Channel	Record Signal	Channel	Record Signal
1	Direct Record (ICS)	15	FM Record (Data)
2	FM Record (Data)	16 1/	FM Record (Data)
3	Direct Record (Digital) Growth Potential	17	FM Record (Data)
4	FM Record (Data)	18	FM Record (Data)
5	FM Record (Data)	19	FM Record (Data)
6	FM Record (Data)	20	FM Record (Data)
7	Direct Record (Servo Reference)	21	FM Record (Data)
8	Direct Record (Servo Reference)	22	FM Record (Data)
9	FM Record (Data)	23	Direct Record (Servo Reference)
10	FM Record (Data)	24	Direct Record (Servo Reference)
11	FM Record (Data)	25	FM Record (Data)
12	FM Record (Data)	26	FM Record (Data)
13	FM Record (Data)	27 1/	Direct Record (Time Code)
14	FM Record (Data)	28	FM Record (Data)

1/ Hardwired playback electronics shall be provided

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b. SLOW/FAST - This dual legend switch shall control electrical selection of tape speed between slow and fast (1-7/8 ips and 7 1/2 ips on standard V2 configuration). Switching shall be interlocked to prevent changing tape speed while the tape is in motion. When recorder power is initially turned on, the fast speed shall be automatically selected.

c. RECORD - Actuation of this switch shall place all channels in the record mode at the selected tape speed, and shall cause the four hardwired reproduce channels and the selected monitor reproduce channel to provide outputs of the data being recorded. Selection of the record mode shall be interlocked to prevent selection of the mode when the tape is in motion.

d. REPLAY - Actuation of this switch shall place the four hardwired reproduce channels and the selectable monitor reproduce channel in the playback mode. Selection of the replay mode shall be interlocked to prevent selection of the mode when the tape is in motion.

e. REWIND - Actuation of this switch shall cause tape motion in the direction from the take-up reel to the supply reel at a speed of approximately 75 ips. Selection of the rewind mode shall be interlocked to prevent selection of the mode when tape is in motion.

f. FORWARD - Actuation of this switch shall cause tape motion in the direction from the supply reel to the take-up reel at a tape speed of approximately 75 ips. Selection of the forward mode shall be interlocked to prevent selection of the mode while the tape is in motion.

g. STOP - Actuation of this switch shall stop tape travel and place the equipment in the ready mode. After a delay of up to 5 seconds from the time the stop mode is initiated, selection of tape speed may be made. Either record, replay, forward, or rewind mode may be selected to place the tape in motion.

h. GO/NO-GO - This dual legend indicator shall be used in conjunction with the BITE mode, as selected by the BITE/NORMAL switch and the channel selector switch, to indicate the operational readiness status of each channel. The reproduced BITE signal is checked by BITE to cause illumination of the appropriate GO or NO-GO legend.

3.4.2.4.2 Other local controls and indicators. The following controls and indicators shall be provided as an integral part of the equipment:

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a. BITE/NORMAL - Successive depressions of this switch shall alternately select the BITE or normal mode. When selected to BITE, this switch shall cause a BITE signal to be routed, via the channel select switch, to each of the 28 channels, as selected one at a time. BITE will provide a GO or NO-GO indication on the GO/NO-GO indicator. A delay of three to five seconds maximum is required for the recorded BITE signal to reach the replay heads and undergo BITE processing. A NO-GO indication appears on the indicator during the delay period. When the BITE/NORMAL switch is selected to normal, it shall permit selection of any one of the 28 channels by the channel select switch for reproduce monitoring.

b. CHANNEL SELECT - This 2-digit thumbwheel switch shall provide a 28 channel selection capability, one channel at a time, and is used in conjunction with both modes of the BITE/NORMAL switch.

c. TACH/TAPE Toggle Switch - In the TACH mode, the tape speed shall be maintained constant by locking the capstan servo drive to the capstan motor tachometer output. The TACH mode shall be automatically selected in the record mode, regardless of the switch position. In the TAPE mode, the capstan servo drive shall be locked to the IRIG 106 servo reference signal recorded on one of two servo reference tracks on the magnetic recording tape. The servo reference track selector switch shall enable selection of track 8 or track 24 for servo control. The switch shall be mounted on the capstan control plug-in card and is visible upon opening the front hinged maintenance cover of the equipment. The Tach/Tape switch shall be visible through the window in the front door of the equipment.

d. TACH INDICATOR - This indicator light shall illuminate whenever the TACH/TAPE switch is placed in either the TACH or TAPE position in the record or replay mode, and when the tape is in motion. The TACH indicator light shall be visible through the glass window in the front door of the equipment.

e. TAPE INDICATOR - This indicator light shall illuminate whenever (1) the TACH/TAPE switch is placed in the tape position, (2) the tape is in motion, and (3) the tape has a prerecorded control track. The tape indicator light shall be visible in the front door of the equipment. In the absence of a prerecorded control signal, servo control shall revert to the tach mode, regardless of the switch position, and the tach indicator light will illuminate.

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f. TAPE COUNTER - This device shall monitor and indicate in four digits the amount of tape, in feet, that is transferred between the supply reel and the take-up reel. The counter may be reset to zero at any time. It shall be visible through the glass window of the front door of the equipment. The counter may be reset by an integral reset control when the front door is open.

g. OVERTEMPERATURE INDICATOR - An indicator light shall be provided to indicate that the maximum safe operating temperature of the equipment has been exceeded. The overtemperature indicator shall be located on the low tape warning sensor so as to be visible when the front door of the equipment is open.

h. SERVO REFERENCE TRACK SELECTOR - A switch shall be provided to permit selection of tracks 8 or 24 to be used for capstan motor speed stabilization in the replay mode. The switch shall be located in a readily accessible position when the front maintenance cover is open.

i. END-OF-TAPE STOP - An end-of-tape (EOT) sensor shall be provided to stop tape travel when the end-of-tape or tape break occurs. This sensing shall be a function of the transport assembly, and shall not depend upon an impulse from the tape or require modification of the tape.

3.4.2.4.3 Remote controls and indicators. The equipment shall have the ability to accept and respond to the following remote control commands and to provide indicator signals to remote indicators.

3.4.2.4.3.1 Momentary remote control commands. The equipment shall accept and respond to the following momentary control signals applied to contacts of the remote control connecter on the rear of the unit. The response to POWER ON and STOP commands shall be immediate. A delay of up to 5 seconds may be required after initiating a STOP command before the equipment will respond to the next command. SLOW, FAST, RECORD, REPLAY, FORWARD, and REWIND commands may only be selected when the equipment is in the stop mode. The momentary command shall be a ground potential applied for a minimum of 500 milliseconds, and shall not be required to exceed 1 second. Local indicator lights and remote indicator signals shall switch to provide the proper mode and status indications. Local and remote momentary commands may be overridden by the steady state commands of 3.4.2.4.3.2

a. POWER Command - The equipment shall accept this command to alternately turn equipment power on or off.

b. SLOW/FAST Command - Each switch actuation shall alternately select the slow tape speed and fast tape speed. The command shall only be accepted in the stop mode.

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c. RECORD Command - Actuation of this switch shall select the record mode. The command shall only be accepted in the stop mode.

d. REPLAY Command - Actuation of this switch shall select the replay mode. The command shall only be accepted in the stop mode.

e. FORWARD Command - Actuation of this switch shall select tape travel in the forward direction at approximately 75 ips tape speed. The command shall only be accepted in the stop mode.

f. REWIND Command - Actuation of this switch shall select tape travel in the reverse direction at approximately 75 ips tape speed. The command shall only be accepted in the stop mode.

g. STOP Command - Actuation of this switch shall stop tape motion.

3.4.2.4.3.2 Steady state remote commands. The equipment shall be capable of accepting and responding to the following external steady state commands applied to appropriate contacts of the remote control connector on the rear of the unit. The command signal shall be either continuously at ground potential or a continuously open circuit. Steady state ground commands shall override their corresponding local or remote momentary command. Steady state record and replay commands (ground potential) shall override a prior stop command, and the equipment shall operate in the mode selected by these steady state commands. When the ground signal is discontinued, the equipment shall return to the stop mode. Indicator lights and remote indication signals shall indicate the proper status and mode.

a. POWER Command - Actuation of this switch shall cause equipment power to be switched on. Return of the switch to the off position shall not cause power to be switched off.

b. SLOW/FAST Command - Actuation of this switch shall command the unit to the slow speed. Returning the switch to the off position shall return the unit to local or remote momentary command control in slow speed. The command shall be accepted in all modes (i.e., even when tape is in motion).

c. RECORD Command - Actuation of this switch shall switch the equipment to the record mode. The command shall be accepted immediately in all modes.

d. REPLAY Command - Actuation of this switch shall switch the equipment to the replay mode. The command shall be accepted immediately in all modes.

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e. TACH/TAPE Command - When a remote control unit having a TACH/TAPE switch is connected to the recorder/reproducer through the remote control connector on the rear of the recorder/reproducer, the tach or tape mode shall be governed by the position of the TACH/TAPE switch on the remote control, regardless of the position of the TACH/TAPE local control switch, except that the tach mode is automatically selected in the record mode.

3.4.2.4.3.3 Remote indicators. The equipment shall have the capacity to operate indicator circuits to denote the status of the equipment and other data as defined herein.

3.4.2.4.3.3.1 Remote status indication. The equipment shall switch ground potential to separate contacts of the remote control external connector on the rear of the unit to indicate each active mode or status. Each of these ground enable circuits shall be capable of supplying an external steady state 80 milliampere lamp load or a 160 milliampere resistive or inductive load, except as noted. These external loads shall be in addition to any internal load these circuits may supply. The following mode and status ground enable outputs shall be provided:

a. Record Mode Active
b. Replay Mode Active
c. Ready Status
d. Power on Status
e. Stop Mode Active
f. Rewind Mode Active
g. Slow Tape Speed
h. Fast Tape Speed
i. Forward (75 ips) Mode Active
j. Low Tape Warning
k. End-of-Tape/Tape Break Status
l. Tach Mode (20 milliampere load from a 5V source)
m. Tape Mode (20 milliampere load from a 5V source)

3.4.2.4.3.3.2 Remote footage counters. Signals shall be provided to separate contacts of the remote control connector to drive a remote footage counter in the forward or reverse direction, such that the remote counter is capable of reading within ±15 feet of the reading on the local footage counter, provided that the remote counter is the same type as the local footage counter.

3.4.2.4.3.3.3 Low tape warning and sequence control. The low tape warning indicates an impending end-of-tape supply. The low tape warning sensor shall provide the ground potential signal to the remote control connector on the rear of the unit when the equipment is in the record or replay mode in either slow or fast speed, and the tape shall continue to run until it has run off the supply reel. When the End-of-Tape/Tape Break sensor arm senses the end-of-tape, the low tape warning ground signal shall cease. In the forward mode, the low tape warning shall cause tape motion to

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stop (actuate stop mode) in addition to providing the ground signal indication. Selection of the forward mode, subsequent to the low tape warning and stop mode, shall initiate a normal forward mode, but shall not inhibit the low tape warning indication. The tape shall travel at approximately 75 ips to the end-of-tape, at which time the equipment shall be returned to the stop mode by the end of tape sensor. When the End-of-Tape/Tape Break sensor arm senses the end-of-tape, the low tape warning ground signal shall cease. The low tape warning sensor shall be adjusted to provide these functions when 110±100 feet of tape is remaining on the supply reel. The low tape warning ground signal shall be capable of operating a remote indicator light and of activating the record mode of another identical recorder/reproducer unit via the remote control connector of that unit. By proper interconnection wiring, multi-machine loop-sequencing shall be possible. In a power down mode, the output of the ground pulse signal line shall provide a high impedance so as not to cause a false indication of sequence activation.

3.5 Detail requirements

3.5.1 Recorder/reproducer AN/AQH-4(V)2 components. The recorder/reproducer shall contain the major components listed in TABLE IV.

TABLE IV. AN/AQH-4(V)2 components.

Components	Quantity
Transport Assembly	1 each
(a) Reel Motor Assembly	1 each
(b) Motor Assembly, Capstan and Tachmometer	1 each
(c) Record Head Assembly	1 each
(d) Reproduce Head Assembly	1 each
(e) Bias Module	1 each
Power Supply Module	1 each
Capstan Servo Module	1 each
Control Module (with BITE)	1 each
FM Record Assembly, 3 channel	1 each
FM/Direct Monitor Reproduce Assembly	1 each
FM Reproduce Assembly, 2 channel	1 each
FM/Time Code Reproduce Assembly	1 each
5 Channel Direct Record Assembly	1 each
Head Box Assembly	1 each
Housing Assembly, Recorder/Reproducer	1 each
Record BITE Channel Select Assembly	1 each
Reproduce Channel Select Assembly	1 each
BITE Switch Assembly	1 each
Counter Module Assembly (Tape Footage)	1 each

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3.5.1.1 Function. This recorder shall record the following: acoustic signals from the aircraft receiver, the intercommunications system, and the ESM receiver; and signals from the IRIG time code generator and the recorder's own internal speed reference. A selectable monitor/reproducer capability shall be part of this device to be used to verify the recording process. There shall be three fixed FM reproduce channels which shall be used in conjunction with the time code and monitor/reproduce channels to allow the replay of prerecorded fixed format training tapes into the aircraft system.

3.5.1.2 Form factor. The Recorder/Reproducer AN/AQH-4(V)2 shall be mechanically interchangeable, except for electrical connectors and for housing, with the AN/AQH-4(V) Recorder/Reproducer. (See FIGURE 1)

3.5.1.3 External finish. The equipment external finish shall be Color Number 36231 in accordance with FED-STD-595.

3.5.2 Performance. Unless otherwise specified herein, values set forth to establish the requisites for required performance shall apply to performance under both standard and extreme service conditions, when in accordance with W-T-001553, magnetic tape is used on precision metal or glass reels supplied to the requirements of W-R-175. Reduced performance beyond the temperature range of 0° and 50°C shall be acceptable due to tape performance limitations only. Unless otherwise specified in the contract, (see 6.2.1) all values of voltage and frequency within the performance limits are allowable for determining full compliance with this specification. The equipment shall also be capable of providing specified performance when semi-precision metal reels are used.

3.5.2.1 Overall performance

3.5.2.1.1 Spectral purity. With reference to a 1 KHz sine wave being recorded at full deviation (±30 percent) and simultaneously being reproduced at peak output level at a given speed, the amplitude of the spectral sidebands and any other frequency components, for any cause, shall be as specified in TABLE V. The table also gives the amplitude of the spectral sidebands due to the dynamic skew and flutter for the case where the following conditions apply: (1) the same 1 KHz sine wave tape is being replayed at the same speed as recorded; (2) the capstan has been rotated from its position relative to the tape during recording to that position which yields the worst case sidebands.

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TABLE V Spectral sideband amplitudes.

Subsequent Replay Speed (Tape Phase-Lock)	Maximum Amplitude Relative to Peak 1 KHz Output Level (Worst Case Capstan Rotation)	
	Simultaneous Reproduced	Subsequent Replay
-45 dB 7-1/2 ips	-35 dB	-30 dB
-32 dB 1-7/8 ips	-22 dB	-17 dB

NOTES: 1. All table values are valid for an ensemble of 16 arithmetic averages.
2. All measurements shall be made with a measurement resolution of 0.15 Hz from 670 Hz to 1330 Hz at either tape speed. For frequencies outside the measurement region, sidebands shall be at least -60 dB below peak output level also, when measured with a resolution of 0.15 Hz.

3.5.2.1.2 Skew

3.5.2.1.2.1 Dynamic skew. The total dynamic skew shall be not greater than ±2.4 microseconds at 7-1/2 ips when measured between adjacent tracks within the same head stack.

3.5.2.1.2.2 Static skew. Static skew shall be in accordance with MIL-STD-1610.

3.5.2.1.3 Crosstalk. The recorded crosstalk between any two tracks of the 28 tracks shall be not greater than that specified in TABLE VI.

TABLE VI Crosstalk.

Crosstalk	Region at 7 1/2	Level
Direct to Direct	500 to 125 KHz	-40 dB
	50 to 500 Hz	-30 dB
FM into Direct	500 to 125 KHz	-40 dB
	50 to 500 Hz	-30 dB
Direct into FM	DC to 31.25 KH	-50 dB
FM into FM	DC to 31.25 KHz	-60 dB

NOTES: 1 Level is relative to Maximum Record Level (DIRECT) or peak deviation of ±30% (FM) and is measured with a 1 Hz resolution in frequency and with an ensemble of 16 arithmetic averages.

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2 All track combinations are subject to this specification requirement.

3 For all the types of crosstalk above, the level of the active channels shall be the appropriate maximum and the channel under test shall be quiescent (shorted input). The levels indicated in the table shall be applicable for all frequencies impressed on the active channels. These levels shall apply to subsequent replay of a tape made on the same recorder/reproducer.

3.5.2.1.4 System polarity. The reproducer polarity shall be equal to the record polarity and shall conform to the requirements of MIL-STD-1610.

3.5.2.1.5 Servo reference frequency. A constant amplitude frequency reference signal shall be provided for recording on the tape on channels 7, 8, 23, and 24. The frequency shall be in accordance with MIL-STD-1610, i.e., 12,500 Hz at 7-1/2 ips, proportional at other speeds. The reference frequency accuracy shall be better than 2 parts in 10^5. The aforementioned characteristics shall be maintained over the temperature range specified in 3.3.10.3.1 and 3.3.10.3.2.

3.5.2.1.6 Cumulative flutter. For tape speeds of 7-1/2 ips and higher, the flutter shall not exceed 0.5 percent peak to peak (two sigma) when measured as directed per IRIG 118 paragraph 3-30, using a MINCOM 8300-W flutter meter or equal. For tape speeds of 1-7/8 and 3-3/4 ips, the flutter shall not exceed 0.6 percent peak to peak (two sigma) when measured in the same manner as prescribed above.

3.5.2.1.7 Time base error. The time base error (TBE) of the equipment, when measured in accordance with IRIG 118, shall be not greater than that indicated in TABLE VII below. The Servo Reference frequency shall be recorded on tape. Upon replay, the Capstan Tachometer/Tape Phase-Locked selector shall be set to TAPE PHASE-LOCKED mode.

TABLE VII Time base error.

Tape Speed	TBE (Peak to Peak in Microseconds)
30	2
15	3
7-1/2	8
3-3/4	14
1-7/8	20

3.5.2.2 Transport assembly. The tape transport assembly shall contain the capstan motor, reeling mechanism, head subassembly, tape guide mechanism, tape footage sensor, elapsed time meters, low tape sensor, and end-of-tape sensor. The following requirements shall be met:

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a. Tape - The tape transport shall accommodate wideband, back coated and non-back coated, one inch wide, 1.0 mil thick, mylar, magnetic tape in accordance with W-T-001553.

b. Reels - The transport assembly shall accommodate 10.5 inch precision reels, of type PCP, in accordance with W-R-175/4; or glass reels in accordance with W-R-175/6. All reels shall be in conformance to W-R-175/4 or W-R-175/6 at the time of testing and use with this equipment.

c. Rewind - Facilities for reversing tape travel and rewinding the tape onto the supply reel shall be provided. The rewind shall have a controlled speed of 75 ips ±10%.

d. Fast forward - Facilities for fast forward movement of the tape from supply reel to take-up reel at a rate of speed equal to the rewind speed shall be provided.

3.5.2.2.1 Record and reproduce heads. Magnetic heads utilized on the transport described herein shall meet or exceed the requirements of MIL-STD-1610 Format A. The following requirements shall be met:

a. Head stack configuration - The magnetic head stack shall be of the rigid block type. All head assemblies shall be of such precision and design that they can be changed in the field without the need for shimming, mechanical adjustment, or alignment other than azimuth adjustment. With such an interchange capability, there shall be no degradation of performance, as specified by the requirements of this equipment specification. The front surface of the head stacks shall present a hard, smooth surface across the entire width of the tape in order to assure a good head-to-tape contact. The heads shall be of the solid ferrite or hard tipped ferrite over metal lamination type and shall not be undercut.

b. Head geometry - There shall be a total of four head stacks, two for recording and two for reproduction. Recording and reproducing head stacks shall be so mounted as to minimize electromagnetic coupling. Each head stack shall contain fourteen individual heads.

c. Head azimuth - The mean gap azimuth shall be not greater than ±1 minute of arc.

d. Head life - The minimum unconditional warranty on the record and reproduction heads, for the combined time of the record and replay modes, shall be 5000 hours, when the equipment is operated at tape speeds of 7-1/2 ips or lower, using tape qualified to W-T-001553. Conditions of warranty shall be as specified in the contract (see 6.2.1).

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e. Head gap - The record head gap shall be 100 ±10 microinches in accordance with MIL-STD-1610. Head manufacturer certified data is acceptable proof of design.

3.5.2.2.2 Reel drive assembly. The reel drive assembly shall be so designed that sufficient tape tension is maintained to allow the system to meet the performance requirements of this specification.

3.5.2.2.3 Tape tension. The equipment shall provide uniform tape tension over the entire reel of tape. The amount of tape tension shall be selected for optimum performance of the equipment when utilizing the tape specified herein. Under no operating conditions shall tape tension exceed the elastic limit for the tape specified.

3.5.2.3 Power supply module. The power supply shall be derived from +28 VDC, in accordance with 3.3.12 thru 3.3.12.3.

3.5.2.3.1 Overload protection. For the purpose of overload protection, the power supply module shall be short-circuit proof and shall not allow an overvoltage condition to occur which would result in a damaging failure to the rest of the recorder/reproducer. For the case where numerous AC or DC voltages are developed, the failure of any one voltage shall not result in additional failure to electronic or solid-state circuitry.

3.5.2.4 Capstan control module. The capstan control module shall provide power to the capstan motor and govern its speed stability, relative to a crystal reference, using a phase-lock servo system. This module shall also provide power to activate the two subminiature indicators which shall be lighted when capstan tachometer phase-lock or tape phase-lock has been achieved. These indicators shall be on the front surface of the module. Visibility of the indicators shall not be obstructed by the tape reels. Indicators may be of the light emitting diode type. This module shall employ a tape phase-locked servo which operates from the servo reference track, 8 or 24 (selectable), in the REPLAY mode. In the event the servo reference signal is not present, the servo shall automatically revert to the capstan tachometer phase-lock mode. A switch shall be provided on the front panel of this module to select phase-lock servo to the capstan tachometer or to enable the servo to phase-lock to the tape reference track (8 or 24).

3.5.2.5 Control and BITE module. The control and BITE module shall provide switching logic for controlling system operation in accordance with 3.3.7.1.7 and 3.4.2.4.2.

3.5.2.6 FM record module (3 channel). The FM record module shall meet the requirement specified in 3.5.2.6.1.

3.5.2.6.1 Concept. The design and construction shall meet with a modular design concept. All FM record modules shall be identical, and electrically and mechanically interchangeable. Each FM record module shall be exchangeable one for another in any one transport and between transports, without necessitating readjustment to meet the requirements specified herein.

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3.5.2.6.2 Contents. The FM amplifier module shall contain three record amplifiers, and shall meet the following requirements:

a. Input level - The FM record amplifier shall be capable of meeting the specified performance with an input voltage of 2.0 volts rms (sine wave). Input levels up to 2.5 volts rms (sine wave) shall not cause damage to the equipment.

b. Input impedance - The FM record amplifier shall provide an input impedance of 20,000 ohms minimum, balanced to ground, with signal ground isolated from chassis ground. A differential input shall be used. Wiring used shall be two wire twisted shielded wires with all shields tied together.

c. Carrier frequencies - FM carrier center frequencies shall be as listed in Table VIII.

TABLE VIII FM carrier center frequencies.

Tape Speed	Center Frequency
1-7/8 ips	14,062 Hz ±3%
7-1/2 ips	56,250 Hz ±3%

NOTE: The FM carrier center frequency for other speeds shall be proportional to the ratio of the speed in question to 7-1/2 ips. The tolerance shall always remain within ±3%.

d. FM signal bandwidth - In accordance with MIL-STD-1610, the FM signal bandwidth shall extend over the frequency range indicated in Table IX for the respective tape speeds. The FM amplifier shall record and reproduce a 38,000 Hz signal of 100 milliseconds duration at a tape speed of 7-1/2 ips. The response shall be no more than 14 dB down from the same level signal at 31,250 Hz for any given amplifier.

TABLE IX. FM signal bandwidth.

Tape Speed	Bandwidth	Center Carrier
1-7/8 ips	DC to 7.82 KHz	14,062 KHz ±3%
7-1/2 ips	DC to 31.25 KHz	56,250 KHz ±3%

Note: The FM bandwidth at other speeds shall be proportional to the speed ratio defined in 3.5.2.6.2(c).

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e. Carrier deviation - The carrier shall be FM modulated with a maximum frequency deviation of ±30 percent, corresponding to peak-to-peak excursions of a 2V rms input signal Peak deviation shall occur at instantaneous peak input level.

f. Deviation direction - An increasing positive voltage shall cause an increasing frequency.

g. Deviation linearity - The linearity of the transfer curve of voltage vs. frequency shall be within ±0.5 percent of the best straight line through zero as measured in accordance with IRIG 118.

3.5.2.7 Direct record module. The direct record module shall contain direct record circuitry for the simultaneous recording of one channel of ICS, one channel of time code data, one channel of digital data (growth potential), and four channels of servo reference frequency data; and shall meet the requirements specified in 3.5.2.7.1 through 3.5.2.7.2.4.

3.5.2.7.1 Concept. The design and construction shall be in keeping with a modular design concept. The direct record module shall be interchangeable between transports without necessitating readjustments to meet the requirements noted herein.

3.5.2.7.2 Contents. The direct record module shall meet the following requirements at tape speeds of 1-7/8 and 7-1/2 ips.

3.5.2.7.2.1 ICS direct record amplifier (one channel)

a. Input Level - the input level shall be between 5 and 12 V rms.

b. Input Impedance - The input impedance shall be 20,000 ohms unbalanced.

c. Signal Bandwidth - The input signal bandwidth for the ICS channel shall be 400 to 3000 Hz.

d. Maximum Record Level - The maximum record level for this channel shall be adjusted such that a 12 V rms, 1 KHz input signal produces a 3rd harmonic distortion $\leq$ 3%.

3.5.2.7.2.2 Time code direct record amplifier (one channel)

a. Input Level - The input signal level shall be between 0.5 and 1.5 V rms and shall be a nominal 1 KHz sine wave modulated in accordance with IRIG Document 104-60 Format B (modified).

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b. Input Impedance - The input impedance shall be 100K ohms (as a minimum) unbalanced.

c. Signal Bandwidth - The time code input signal bandwidth shall be between 60 Hz and 9000 Hz.

d. Maximum Record Level - The maximum record level for this channel shall be adjusted for optimum performance with any TD-900A Time code generator.

3.5.2.7.2.3 Servo reference direct record amplifier (four channel). Each servo reference record amplifier shall derive its reference frequency from a common source. The reference frequency recorded from all 4 channels simultaneously shall be 3,125 Hz ±0.002% at 1-7/8 ips, 12,500 Hz ±0.002% at 7-1/2 ips and the established proportion at other tape speeds.

3.5.2.7.2.4 Digital annotation direction record amplifier (growth potential only).

3.5.2.8 Reproduce modules. There shall be three, two-speed reproduce modules as specified below:

a.	FM Reproduce Module - (2 Channel)	Tracks 12 and 14
b.	FM/Time Code Reproduce Module -	Tracks 16 and 27
c.	FM/Direct Monitor Reproduce Module	Any of 28 tracks selectable one at at time

3.5.2.8.1 FM reproduce module (two channel). This two channel FM reproduce module shall meet the requirements specified in 3.5.2.8.1.1 and 3.5.2.8.1.2.

3.5.2.8.1.1 Concept. The design and construction shall be in keeping with a modular design concept. This module shall be electrically and mechanically interchangeable between transports without necessitating readjustments to meet the requirements specified herein.

3.5.2.8.1.2 Contents. This module shall contain two-speed electronics for simultaneous reproduce or subsequent replay of FM data recorded on tracks 12 and 14 at 1-7/8 and 7-1/2 ips within the performance limits specified below:

a. Frequency Response - The frequency response shall extend over the bandwidth as indicated in Table X for the respective tape speeds. The FM response shall be uniform within ±0.5 dB over the frequency range of DC to 2500 Hz and ± 2.0 dB from 2500 Hz to the high frequency shown for each tape speed. Measurement shall be based upon 0 dB reference level at 1000 Hz.

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TABLE X Frequency response.

Tape Speed	Bandwidth FM Monitor
1-7/8	DC to 7.83 KHz
7-1/2	DC to 31.25 KHz

b. FM Signal to Noise Ratio - The signal to noise ratio (SNR) for an output signal level of 2.0 volts rms shall be 65 dB per one hertz bandwidth and 31 dB broadband at 7-1/2 ips. The SNRs shall be 50 dB per one hertz bandwidth and 28 dB broadband at 1-7/8 ips. The noise floor shall be -75 dB per one hertz bandwidth for 7-1/2 ips excluding discretes; and 70 dB per one hertz bandwidth for 1-7/8 ips excluding discretes, up to 6999 Hz, and -65 dB per one hertz bandwidth for 1-7/8 ips from 7000 to 7820 Hz, with respect to the amplitude corresponding to peak deviation. An external band limiting filter, having a cut off characteristic of 24 dB/octave and a cutoff set at the upper band limit, may be used in the measurement of broadband SNR.

c. FM Phase Linearity - Deviation from the best straight line shall be within ±3.0 degrees over the recorded frequency band from 0.1 Hz to 18,000 Hz. From 12,000 Hz to 18,000 Hz, deviation shall be within ±0.5 degrees. This specification applies to the FM electronics only and does not include the use of tape. A Hewlett Packard Gain/Phase Meter type 3575A or equal shall be used to determine this linearity.

d. FM Amplitude Linearity - For the reproduce process only, the amplitude transfer function linearity shall be ±0.25 percent of peak-to-peak deviation from a best straight line through zero.

e. Harmonic Distortion - All FM harmonic distortion components shall be a minimum of 40 dB below a peak deviated signal recorded at 0.25 band edge for 7.5 ips (36.5 dB for 1-7/8 ips). An external band limiting filter, as described in (b) above, shall also be used for measuring harmonic distortion. All intermodulation distortion components shall be a minimum of 45 dB below peak deviation for any and all frequency combinations and amplitude combinations whose aggregate amplitude causes peak deviation of the FM carrier.

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f. Differential Time Delay - The differential time delay of the output signals between any two FM reproduce channels shall be not greater than 250 microseconds when tested with a common input to the two channels, in the frequency range 200 to 5000 Hz.

g. FM Output Level - The output of each FM reproduce amplifier when delivered to a parallel load of 2000 ohms (min) and 200 pf shall be 2 V rms ±10 percent for a 1 KHz, 2V rms input signal recorded at 1-7/8 and 7-1/2 ips, when simultaneously or subsequently reproduced at these tape speeds.

3.5.2.8.2 FM/time code reproduce module. This two channel reproduce module shall meet the requirements specified in 3.5.2.8.2.1 through 3.5.2.8.2.3.

3.5.2.8.2.1 Concept. The FM/time code reproduce module shall be electrically and mechanically interchangeable between transports without necessitating readjustments to meet the requirements specified herein. Time code signals shall be recordable on tracks 16 (FM) and 27 (Direct), and shall be reproducible at tape speeds of 1-7/8 and 7-1/2 ips.

3.5.2.8.2.2 Contents. This module shall contain a two-speed FM reproduce amplifier for track 16 and a direct reproduce amplifier for track 27. The FM reproduce amplifier shall meet the reproduce performance parameters specified for the two channel FM reproduce module in 3.5.2.8.1. The direct reproduce amplifier shall contain the necessary circuitry for simultaneous reproduction or subsequent replay of time code generator data recorded on track 27 at 1-7/8 and 7-1/2 ips. The following performance limits shall be met when this time code data or equivalent is continuously reproduced at tape speeds between 1-7/8 and 75 ips. This direct reproduce module shall be energized in all modes of operation, including stop.

3.5.2.8.2.3 Time code data. Time code data recorded from a TD-900A or equal on track 27 shall be simultaneously or subsequently reproduced by this direct reproduce amplifier over the speed range from 1-7/8 and 75 ips. In addition to the foregoing requirement, the performance parameters listed in Tables XI and XII shall be met.

3.5.2.8.3 FM/direct monitor reproduce module. The FM/direct monitor reproduce module shall meet the requirements specified in 3.5.2.8.3.1 and 3.5.2.8.3.2.

3.5.2.8 3.1 Concept. The design and construction shall be in keeping with a modular design concept. The reproduce monitor shall be capable of switching each of the 28 reproduce tracks one at a time to either an FM reproduce monitor amplifier or direct reproduce monitor amplifier according to the type of data recorded on the track. The output of each of the monitor reproduce amplifiers shall be provided to separate pins of the

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signal output connector. Each monitor amplifier shall be capable of operation at 1-7/8 and 7-1/2 ips. Each monitor reproduce module shall be exchangeable between transports without necessitating readjustments to meet the requirements specified herein.

TABLE XI. Time code data frequency response.

RECORD SPEED	REPRODUCED SPEED		
	1-7/8 IPS	7-1/2 IPS	75 IPS
1-7/8 IPS	+3dB 60Hz-300Hz -5dB +3dB 300Hz-3000Hz -6dB	+3dB 60Hz-300Hz -5dB +3dB 300Hz-9000Hz -3dB	+3dB 2400Hz-60Hz -6dB
7-1/2 IPS		+3dB 60Hz-300Hz -5dB +3dB 300Hz-9000Hz -3dB	+3dB 600Hz-15KHz -6dB

NOTE: Frequency response shall be referenced to 1KHz output real time or proportional frequency output at other speeds.

TABLE XII Time code data output levels.

RECORD SPEED	REPRODUCED SPEED		
	1-7/8 IPS	7-1/2 IPS	75 IPS
1-7/8 IPS	1.7V rms +3.5dB -5.5dB	1.7V rms +3.5dB -5.5dB	1.7V rms +3.5dB -5.5dB
7-1/2 IPS	1.7V rms +3.5dB -5.5dB	1.7V rms +3.5dB -5.5dB	1.7V rms +3.5dB -5.5dB

NOTES: 1 The output levels shall be in accordance with this table for an input level of 1 V/rms at 1 KHz and an output load of 600 ohms.

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2 The reproduced frequency is proportional to reproduce/record speed ratio.

3 Signal to Noise - The SNR of a 1 KHz, 1 V/rms signal recorded at 1-7/8 and 7-1/2 ips shall be greater than 26dB when reproduced at 1-7/8, 7-1/2 and 75 ips (reproduced frequency proportionate to tape speed).

4 Harmonic Distortion - The third harmonic distortion of a 1 KHz 1 V/rms input signal recorded at 1-7/8 and 7-1/2 ips shall be sufficient to provide optimum performance with any properly operating TD-900A Time Code generator. For a 1KHz reference signal recorded at 1-7/8 or 7-1/2 ips, the third harmonic distortion shall not exceed 3 percent when reproduced at the same tape speed.

3.5.2.8.3.2 Contents. This module shall contain a two-speed reproduce amplifier (1-7/8 and 7-1/2 ips), an FM filter for each speed and a direct equalizer for each speed. This module shall meet the following requirements:

a. FM Performance Limits - When selected to any of the 21 FM tracks listed in 3.4.2.3, the reproduce performance limits shall be as specified in 3.5.2.8.1.2.

b. Time Code Performance Limits (Track-27) - Monitor only. For performance limit checks, use TK-27 hardwired output (see 3.5.2.8.2.2).

c. ICS Performance Limits (Track-1) - When selected to this track, the direct reproduce amplifier shall reproduce signals within ±3 dB, over the frequency band of 400 to 3000 Hz, with 0 dB, reference level at 1000 Hz. The 3rd harmonic of a 1 KHz, 12 V/rms signal reproduced at either speed shall be less than 3 percent. The reproduced output level of a 1KHz, 12 V/rms input signal, recorded at 1-7/8 or 7-1/2 ips, shall be 3 V/rms ±3 dB. The same 4:1 reduction rate shall apply to any 1 KHz input signal between 5 and 12 V/rms, i.e. 5 V/rms input = 1.25 V/rms output (±3 dB).

d. Servo Reference Performance Limits (Track 7, 8, 23, 24) - When tracks 7, 8, 23, or 24 are selected for replay, the requirements specified in Table XIII shall be met when locked to tape. In addition, the flutter requirements specified in 3.5.2.1.6 shall be met for each of these four channels, when they are reproduced at 1-7/8 and 7-1/2 ips.

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TABLE XIII Replay servo reference performance limits.

Tape Speed	Replay Servo Reference Frequency
1-7/8	3,125 Hz ±0.002%
7-1/2	12,500 Hz ±0.002%

e. Digital Channel - Growth potential

3.5.2.9 Bias/reference module. The bias reference module shall provide the sine wave bias recording frequency of 2 MHz or greater and the servo reference frequencies specified in 3.5.2.1.5.

3.5.2.10 Main frame. The main frame of the recorder/reproducer shall provide mechanical and electrical connections to the transport assembly and all the modules specified in 3.5.1. The main frame shall incorporate handles, as an integral part, reel covers, and a footage counter.

3.5.2.10.1 Reel cover. A cover shall be provided to enclose the reels, the tape supply, and the tape footage counter. It shall not be necessary to open or remove the cover from the transport to view the tape footage counter. The tape threading path and any special operating instructions shall be permanently marked on the interior surface of the reel cover. The reel cover shall provide for convection cooling of the reel and drive assembly. The technique selected for this cooling shall be consistent with 3.3.6 for interference control and shall be so designed as to minimize dirt/dust entry to the tape/reels and drive areas.

3.5 2.10.2 Tape footage counter. A tape footage counter shall be provided. The following requirements shall be met:

a. Resolution - The counter shall have a four-digit display.

b. Accuracy - The counter accuracy shall be ±5 percent or one count, whichever is greater.

3.6 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

3.6.1 Soldering. Soldering shall be performed in accordance with MIL-STD-1460.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, or purchase order, the contractor may use his own or any other facilities suitable for the performance of the

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inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (see 4.2)

b. Initial production tests. (see 4.3)

c. Acceptance tests. (see 4.4)

d. Maintainability verification. (see 4.5)

e. Reliability verification test. (see 4.6)

4.2 First article inspection. First article inspection shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. First article inspection shall be accomplished under the approved test procedure of 4.6. The government inspector and the procuring activity shall be advised when inspection or tests are to be conducted, so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the government.

4.2.1 First article inspection data. The contractor shall include all data collected in conducting these tests for procuring activity review. (see 6.2.2)

4.2.2 Scope of tests. First article inspection shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications, and the contract. First article inspection shall include: environmental tests in accordance with the procedures of MIL-T-5422; electromagnetic interference tests in accordance with the procedures of MIL-STD-462; and power input tests in accordance with the procedures of MIL-STD-704.

4.2.2.1 Environmental tests. Environmental tests shall be conducted in the sequence specified in MIL-T-5422. Environmental tests shall include, but not be limited to, the requirements specified in the 4.2.2.1.1 through 4.2.2.1.4.

4.2.2.1.1 Vibration test. The equipment shall be vibrated at random for 10 minutes, 6 GRMS, at a frequency between five and zero Hz for 10

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minutes during each hour of equipment operating time. The direction of vibration shall be vertical to the normal mounting plane.

4.2.2.1.2 Temperature/altitude tests. A temperature/altitude test shall be performed in accordance with MIL-T-5422.

4.2.2.1.3 Shock test. A shock test shall be performed in accordance with MIL-T-5422.

4.2.2.1.4 Humidity test. A humidity test shall be performed in accordance with MIL-T-5422.

4.2.2.2 Electromagnetic interference emission and susceptibility tests. The required tests and test limits are defined in MIL-STD-461 for Class 1C equipment. The techniques to be used for the measurement and determination of the electromagnetic interference characteristics (emission and susceptibility) are defined in MIL-STD-462.

4.2.2.2.1 MIL-STD 461 and MIL-STD-462 modifications.

a. CS 01 - This test shall be performed on all power lines to the unit under test.

b. CS 02 - A 10 microhenry choke shall be inserted in series with the power line under test at the point where the Line Stabilization Capacitor was removed. Applicable Limits: Monitoring device, VTVM or Oscilloscope, shall read a minimum of 1 Volt.

c. CS 06 - The waveshapes (spikes) of Figures 6 and 7, herein shall be used.

d. RS 02 - This test shall be performed on the power line also. Applicable limits:

(1) Power Frequency Test - Twenty amperes shall be applied while the frequency is slowly varied from 390 Hz to 410 Hz.
(2) Spike Test - The waveshapes (spikes) of Figures 6 and 7, herein shall be used.

e. RS 03 - The following testing limits shall be used:

	Frequency	Field Strength
(1)	14 KHz to 2 MHz	10V/M
(2)	2 MHz to 30 MHz	40V/M
(3)	30 MHz to 100 MHz	5V/M
(4)	100 MHz to 400 MHz	10V/M
(5)	400 MHz to 10 GHz	5V/M

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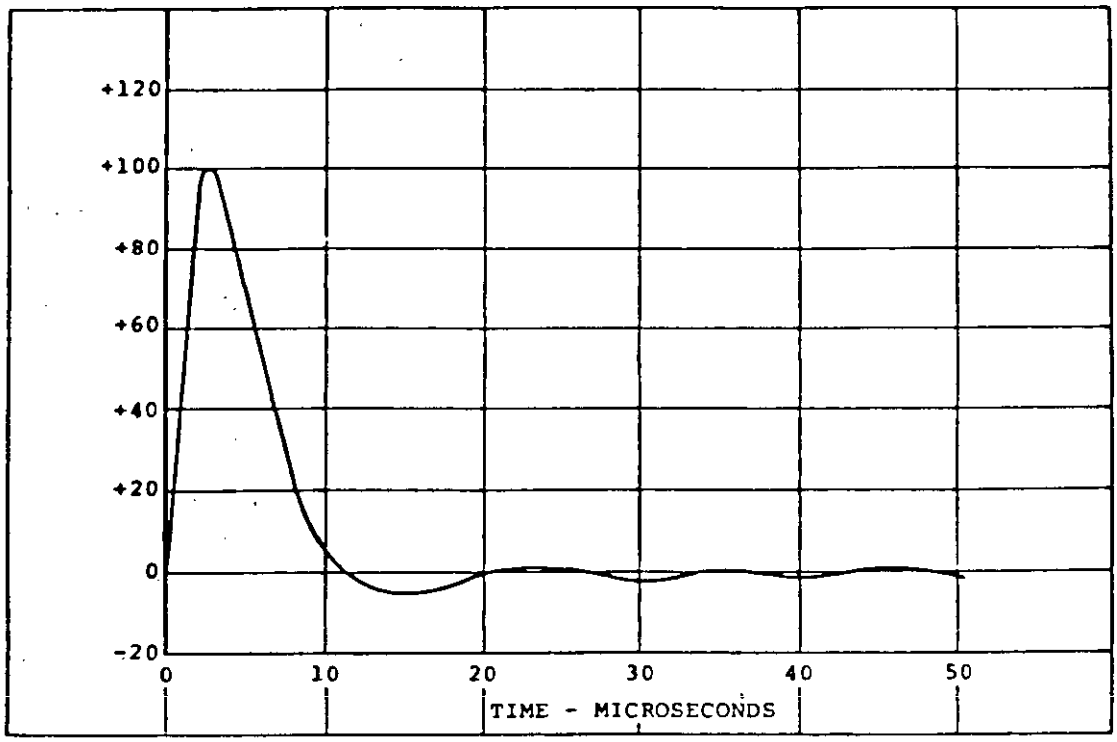


Figure 6 Long Pulse characteristics.

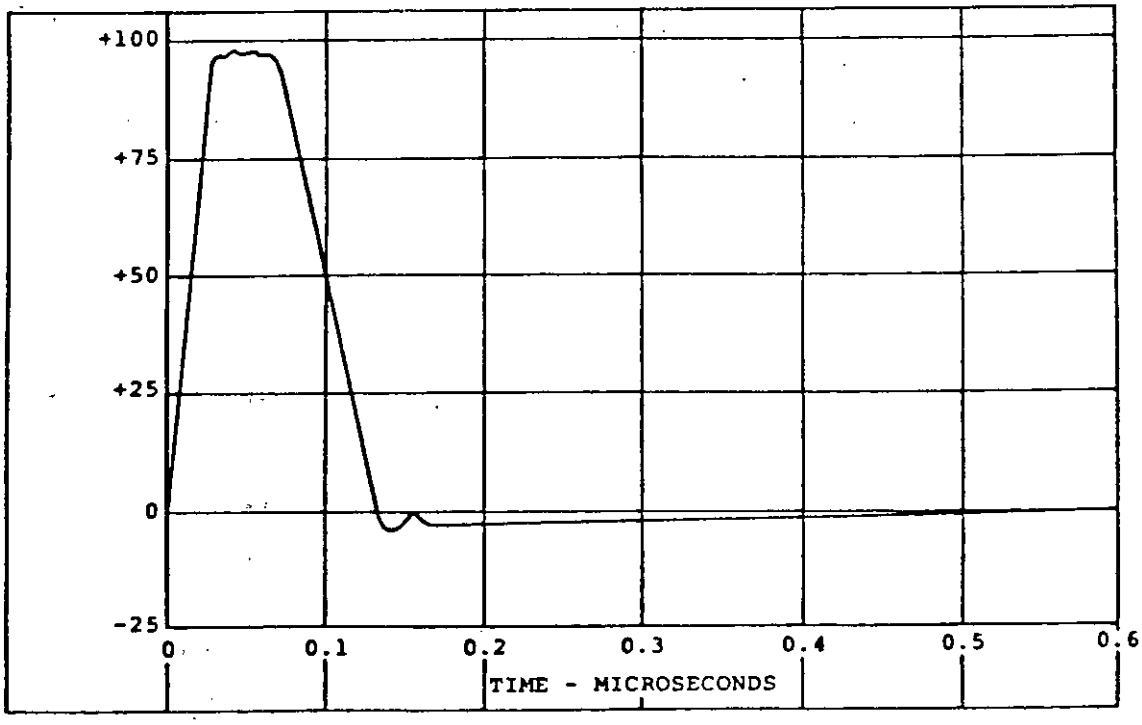


Figure 7 Short pulse characteristics.

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f. CE 04 - Limit at 4 MHz only is relaxed to 39 dB $>$1μA in lieu of specification limit of 20 dB $>$1μA for measurement on the combined cables from J3 and J4 external connectors.

g. RE 02 - Is relaxed to 27 dB $>$1μV/M at 4 MHz only, in lieu of the specification limit of 24 $>$ dBμ1 V/M.

4.2.2.3 Power input test. Each qualification test sample equipment shall be subjected to a range of input voltages as specified herein:

a. The input voltage shall be varied within the limits of Figure 9 of MIL-STD-704 as specified in 3.3.12. It shall be demonstrated that the equipment will perform as required by 3.4.

b. It shall be demonstrated that input voltage transients specified above will not cause damage to the equipment, and when the voltage returns within the specified limits normal operation shall automatically resume.

4.2.2.4 Maintainability demonstration tests. A maintainability demonstration test shall be conducted in accordance with MIL-STD-471, to demonstrate equipment compliance with the specified MTTR and turn-around time requirements. In addition, the test shall demonstrate that: the BIT function and self-test features are capable of fault detection and isolation to a WRA; repair is accomplished through removal and replacement; and a repair verification is accomplished by exercising the stated equipment BIT and self-test features.

4.2.2.5 Reliability test. A reliability development test (RDT) shall be conducted in accordance with MIL-STD-2068. The equipment(s) under test shall be representative of future production equipments in materials, configuration and workmanship. The test is intended to identify, analyze, and correct all deficiencies. All equipment failures shall require analysis and corrective action (see 6.2 2).

4.2.2.5.1 RDT failure reporting analysis and corrective action. All RDT failures shall be reported to the procuring activity cognizant engineer or his appointed designate within 24 hours of the failure incident (see 6.2.2). All failures shall be analyzed. Positive corrective actions shall be developed for failures. Necessary modifications shall be incorporated into RDT equipments under test for the purpose of corrective action verification. The procuring activity may require verification of the corrective action modification by testing at a lower level of assembly or at the piece part level, as deemed necessary.

4.2.2.5.2 RDT corrective action. Verified corrective action(s) shall be incorporated in all systems submitted for acceptance, and delivered systems shall be retrofitted.

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4.2.3 First article sample approval. Approval of the first article sample will be given by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the first article sample. Prefabrication of production equipment prior to the approval of the first article sample is at the contractor's own risk.

4.2.4 Production equipment. Equipments supplied under the contract shall, in all respects, including design, construction, workmanship, performance and quality, be equal to the approved first article sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection; for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as acceptable to the procuring activity.

4.3 Initial production test. One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment will be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The first article sample shall not be selected for this test.

4.3.1 Scope of tests. This equipment shall be subjected to any and all tests the procuring activity deems necessary to ensure the production equipment is equal to the previously approved first article sample in design, construction, workmanship, performance and quality, and to ensure that it meets all applicable requirements.

4.3.2 Accessory material. In addition to the complete equipment subjected for Initial Production Tests, the contractor shall also provide accessory material and include data necessary to test the equipment.

4.3.3 Initial production sample approval. Approval of the Initial Production Sample will be by the procuring activity upon satisfactory completion of all tests. Any design, material, or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.4 Acceptance tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Representative. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall maintain test reports showing

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quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual Tests. (see 4.4.1)

b. Sampling Tests. (see 4.4.2)

4.4.1 Individual tests. Each equipment submitted for acceptance shall be subjected to individual tests. These tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy, and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Product. (see 4.4.1.1)

b. Operational Test. (see 4.4.1.2)

c. Manufacturing Burn-In Test. (see 4.4.1.3)

4.4.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure required operation of the equipment.

4.4.1.3 Manufacturing burn-in test. The contractor shall perform burn-in processing of each equipment, which shall include 80 hours of equipment operating time. Temperature cycling shall be over the temperature range of -55°C to +71°C, with dummy tape inserted and equipment power applied during the heating cycle. After equipment stabilization at +55°C, electrical checks shall be conducted on each of the 28 channels.

4.4.1.3.1 Vibration test. The equipment shall be vibrated at random for 10 minutes, 6 GRMS, at a frequency between five and zero Hz for 10 minutes during each hour of equipment operating time. The direction of vibration shall be cycled through the axis as listed below:

a. Vertical to the normal mounting plane.

4.4.1.3.2 Burn-in procedure. Manufacturing burn-in shall be in accordance with the procuring activities approved contractor burn-in procedure.

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4.4.1.3.3 Failure. Should a failure occur during burn-in, processing shall be interrupted, the equipment restored to operating condition, and the procedure continued.

4.4.1.3.4 Failure cycles. The last two cycles shall be failure free. The definition of a complete temperature cycle shall be acceptable to the procuring activity.

4.4.2 Sampling tests. Equipments selected for sampling tests shall have first passed the individual tests. Equipment will be selected for sampling tests by the Government representative in accordance with Table XIV.

TABLE XIV. Sampling requirements

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1 1/
Next 50	1
Next 75	1
Next 100	1 2/

1/ One out of first ten need not be selected and tested if initial production tests are conducted. Sampling tests are not required when reliability assurance tests are conducted.

2/ for each additional 200 or fraction thereof

4.4.2.1 Scope of tests. As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the first article, initial production, individual, and special tests.

c. Manufacturing burn-in test specified in 4.4.1.3, except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

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4.4.3 Reliability assurance test. A Reliability assurance test shall be conducted in accordance with MIL-STD-781. Tests as required by both the qualification phase and the production acceptance (Sampling) phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and MIL-STD-2074.

4.4.3.1 Qualification phase. When specified in the contract (see 6.2.1) and prior to the acceptance of equipments under the contract or order, a minimum of three equipments shall be tested as outlined in MIL-STD-781. The maximum number of equipments to be used shall be 10. For the qualification phase, a fixed-length test with a discrimination ratio of 2.0 and an 80 percent confidence level shall be performed. The specified system MTBF is defined as 90 hours. Test conditions shall be in accordance with the requirements of MIL-STD-781.

4.4.3.2 Reliability production acceptance (sampling) phase test. The equipment, throughout production, shall be tested as specified in MIL-STD-781 (as modified herein). except that all equipments produced shall be tested.

4.4.3.2.1 All-equipment test. Each ATR produced shall be subjected to an all-equipment test consisting of the following environments:

a. Random vibration. (see 4.4.3.2.1.1)

b. Thermal cycling. (see 4.4.3.2.1.2)

These environments may be applied sequentially in the order listed. Test levels, durations, and tolerances shall be as specified in MIL-STD-781 for optional, nonstatistical production acceptance reliability tests, and as specified in 4.4.3.2.1.1 through 4.4.3.2.1.3. Following random vibration, each equipment produced shall be tested for a minimum of 50 operating hours in accordance with the conditions specified herein. All failures shall be repaired and the test continued except that the last 10 hours of operation shall be failure free. A failure shall be defined as any out-of-specification performance or any condition which requires repair of the equipment under test. Adjustment of the controls normally accessible to an operator shall be permitted.

4.4.3.2.1.1 Random vibration. Testing of more than one axis may be performed. If exercised, the testing shall be in accordance with the approved test procedure.

4.4.3.2.1.2 Thermal cycling. Each cycle shall consist of lowering the chamber temperature to -30°C, with equipment nonoperating until the equipment temperature stabilizes. During the cold period the equipment shall be turned on and tested, then the chamber temperature shall be raised to +55°C. The equipment shall be tested at the high temperature.

4.4.3.2.1.3 Reject/accept criteria. The reject/accept criteria shall conform to the requirements of the all-equipment production reliability

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acceptance test plan of MIL-STD-781. During the all-equipment test, each equipment shall be exercised and tested in accordance with the approved test procedure. Should a failure occur, the equipment shall be repaired, proper operation verified, and the test continued commencing at the beginning of the cycle in which the failure occurred. Corrective action shall be taken, should failures occur. Records shall be kept of all failures. To determine whether a reject condition has been reached at any time during the contract, the operating test hours and the failures therein shall be totaled, and the results compared with the all-equipment production reliability acceptance test plan of MIL-STD-781. Any time that the current total of test and test failures shows a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment any time that a reject situation exists, pending a review of the Contractor's efforts to improve the equipment. Any corrective actions made shall be incorporated in all units at the contractor's expense. All equipment shall be subjected to, and shall pass, the individual tests prior to the all-equipment test. When repair to an equipment during or after all-equipment testing is accomplished by card substitution, that card shall have been subjected to the burn-in requirements for spares as specified in 4.4.3.4.

4.4.3.3 Test detail. The test details, such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, etc., shall be part of the test procedures acceptable to the procuring activity prior to the beginning of the reliability assurance tests.

4.4.3.4 Burn-in requirement for spares. The reliability testing procedures specified below shall be applied to all assemblies (SRAs and sub-SRAs) that are procured separately.

a. As a minimum, in-process burn-in, screening, and other individual testing of separately procured assemblies shall be equivalent to that conducted on production assemblies. Following the burn-in, a complete functional performance test shall be conducted on each assembly offered for acceptance.

b. Included in the testing of each assembly shall be at least 40 hours of power-on operation (signal voltages not required) under the environmental test conditions specified in 4.4.1.3. Power supply assembly burn-in shall be conducted with the power supply loaded between 80 - 100 percent of rated full load.

c. The tests specified herein shall be conducted by, or for, the supplier of the assemblies. Assemblies shall not be accepted by contractors for acceptance by the or a government without objective evidence of having passed the above and other required tests. These test requirements

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shall be included in the manufacturing drawing, specification, or source controlled drawing for the assemblies.

4.4.4 Special tests. When specified in the contract, (see 6.2.1) special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment, and to assure adequate quality control. The equipment for special tests may be selected from equipment previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special test schedule. Selection of equipments for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required (this will be determined by the procuring activity).

4.4.4.2 Scope of tests. Special tests shall consist of such tests as are acceptable to the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable. When these previously approved tests are not applicable, the contractor shall prepare a test procedure for procuring activity acceptance prior to conducting the tests (see 6.2.2).

4.4.5 Equipment failures. Should a failure occur during either the sampling, reliability assurance, or special tests, the following action shall be taken:

a. Determine the cause of the failure.

b. Determine if the failure is an isolated case or design defect.

c. Include for procuring activity acceptance, proposed corrective action intended to reduce the possibility of the same failure(s) occuring in future tests. (see 6.2.2)

d. Where practical, include a test in the individual test to check all equipment for this requirement, until reasonable assurance is obtained that the defect has been corrected.

4.5 Maintainability verification. A maintainability verification demonstration shall be performed to verify that the requirements of 3.3.7 thru 3.3.7.1.9 have been met in accordance with maintainability demonstration procedures which have been previously acceptable to the procuring activity.

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4.6 Reliability verification test. Reliability verification tests shall be conducted in accordance with the test plan specified below. All failures occuring during testing shall be analyzed by the contractor for cause. Any failure determined jointly by the contractor and procuring activity to be in violation of the equipment specification or other contractual documents shall be corrected at contractor expense and fixes incorporated in all production units. Such fixes shall be validated by subsequent testing.

4.6.1 Objectives. Perform reliability testing that shall:

a. Stress the recorder to the point of determining its primary failure modes, without exceeding maximum permissible stress on parts.

b. Environmentally cycle test the recorder to afford a specified confidence that it will meet its specified MTBF.

4.6.2 Verification requirements.

a. Testing shall be limited to thirteen weeks at an approximate rate of 80 hours "on" time per week, using two Units Under Test (UUT).

b. "Step-Stress" tests shall be run for six weeks.

c. Cycling tests shall be run for seven weeks.

d. A procuring activity engineer will be assigned to the test facility to observe reliability testing.

e. Test procedures and reports shall be prepared by the contractor (see 6.2.2).

4.6.3 Test requirements.

4.6.3.1 Step-stress test.

a. A thermal survey shall have been conducted prior to this test. It shall have identified 15 parts that first approach their maximum thermal ratings as ambient temperature is raised. This survey shall also monitor tape path components (head, rollers, etc.) to determine at which ambient temperature these components reach the magnetic tape's maximum operating limits.

b. As part of this test program, and as immediate prelude to the step-stress test, an 80 hour burn-in shall be conducted to filter out infant mortalities. Corrective action shall be taken for any "pattern" failures.

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4.6.3.1.1 Failure definition. A failure shall be defined either as: (1) a catastrophic condition, where the unit no longer operates; or (2) a sub-standard condition where the unit continues to operate, but out of tolerance readings are observed.

4.6.3.2 Step-stress test procedure.

PHASE I

a. Immediately following the burn-in processing and beginning at an ambient temperature of +25°C, the temperature shall be increased in increments of 5°C. The UUT shall be loaded with the tape recommended by the contractor for normal operation. Thirty minutes after the chamber has reached the first temperature level, the UUT shall be considered to be temperature stabilized and a functional test shall be run on the UUT to evaluate its performance.

b. After successfully completing the functional test, the ambient temperature shall be increased in successive 5°C increments. Thirty minutes after the chamber has reached its latest temperature increment, it shall be considered to be temperature stabilized, and a functional test shall be run on the UUT to evaluate its performance.

c. When a catastrophic failure occurs, testing shall be discontinued, the chamber brought to room ambient temperature, the equipment repaired, the chamber returned to the temperature at which the failure occurred, and the test resumed.

Phase I tests shall be terminated when the maximum case temperature of electrical components, as previously determined during the temperature survey, is reached. When tape path components approach the limit at which the tape should be operated, the tape shall be replaced by a dummy tape, and the tests shall be continued.

PHASE II

a. With the dummy tape loaded, temperature shall again be increased in increments of 5°C with the UUT allowed to stabilize, as in Phase I; the equipment shall be run through its various modes of operation and monitored for catastrophic failure (the first step of Phase II shall commence at the last temperature level reached during Phase I).

b. Chamber temperature shall be increased until a failure occurs or until one of the parts being monitored reaches a case temperature within 15 percent of its manufacturer's rating; this shall be considered its operational limit. Failures shall be dealt with as described under Phase I. When a part's "limit" is reached before the chamber temperature has reached the specification limit of +55°C, corrective active is mandatory, even though no failure has occurred. Corrective action may be in the form of improved head

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sinking, selection of a new part, or relocation of the "limited" part. When the part's "limit" is reached at a chamber temperature beyond +55°C, the procuring activity and the contractor shall determine whether it is prudent to take corrective action (for example, where one part reaches its limit long before any of the others do).

c. Step-stress testing shall be terminated at a temperature level where all reasonable corrective measures have been taken to maximize the chamber temperature at which the 15 critical parts reach their limit at about the same time.

4.6.3.3 Cycling tests.

a. The UUT shall be cycled from ambient temperature to +55°C at a rate of 5°C per minute or greater and maintained at that temperature for ten hours. During that time, the UUT shall be functionally cycled through all its modes of operation, and BITE checks shall be conducted twice (as an example, at the 4th and 8th hour). The UUT shall be vibrated at a frequency between 20 and 60 Hz with peak acceleration of 6 GRMS random for ten minutes during each hour of equipment operating time. The BITE functions shall be exercised one time during one of these vibration periods.

b. The UUT shall then be turned off, the temperature reduced to ambient, the tape replaced by a dummy tape, and the chamber cooled to -54°C and held there for four hours.

c. At the end of this period, the chamber shall be returned to +25°C, and stabilized. A tape shall then be loaded and a functional test performed. After successful completion, the cycle shall be reinitiated by heating the chamber up to +55°C again.

d. This process shall be repeated continuously until seven weeks pass or 500 operating hours are completed, whichever occurs last.

e. All failures shall be analyzed by the contractor for cause, and remedial measures shall be implemented where deemed necessary by the procuring activity.

4.7 Presubmission Testing. No item, part, or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements of this specification and the contract.

5. PACKAGING

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5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or purchase order in accordance with MIL-E-17555 or MIL-STD-794 as applicable.

6. NOTES

6.1 Intended Use. The equipment covered by this specification is intended to provide a 28 - channel airborne magnetic tape recording/reproducting system suitable for ASW data recording, storage and reproduction. The equipment specified herein will be capable of accomplishing this function through the use of direct recording and wideband FM recording techniques as specified in this document. The equipment is intended for use in ASW aircraft to record SONAR, LOFAR, DIFAR, ESM, ICS, Digital Data, and Time Code information.

6.2 Ordering Data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Performance requirements if other than specified in 3.4 and 3.5.2.

c. Conditions of warranty (see 3.5.2.2.1d).

d. Selection of applicable levels of testing (see 4.1.1).

e. Corrective action on previously accepted requirement, if required (see 4.3.3).

f. Qualification phase if required (see 4.4.3.1).

g. Special tests, if required (see 4.4.4).

h. Selection of applicable levels of packaging and packing (see 5.1).

6.2.2 Data Requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

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Paragraph no.	Data requirement title	Applicable DID no.	Option
3.2.1	Nonstandard part approval request	DI-E-7028A	
3.2.2	Nonrepairable assemblies approval	UDI-L-26529	
3.2.4	Semiconductor devices approval	UDI-E-21337	
3.3.2	Plan, reliability program	DI-R-7079	
3.3.2.4.1	Reliability guarantee	DI-R-7035	
3.3.4	Drawing requirements for rack or console	UDI-E-21339	
3.3.8	Nomenclature and identification assignment and approval request	DI-E-21336B	
4.2.1	First article test report	UDI-T-23790	
4.2.2.5	Reliability test procedure	UDI-T-21347	
4.2.2.5.1	Reliability test and demonstration reports	DI-R-7034	
4.4.1.3.1	Test procedures	UDI-T-21347	
4.4.4.2	Test procedures, special tests	DI-T-2187	
4.4.5	Corrective action report	DI-R-1704	
4.6.2(e)	Test Procedures	UDI-T-21347	
4.6.2(e)	Test Reports	DI-T-2072	

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer).

6.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence should apply:

a. Contract - The contract should have precedence over any specification.

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b. This specification - This specification should have precedence over all applicable subsidiary specifications.

c. Reference specifications - Any referenced specification (Section 2.) should have precedence over all applicable subsidary specifications referenced therein. All referenced specifications should apply to the extent specified.

6.4 Performance objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which should be considered in the production of this equipment. Where it appears a substantial reduction in size and weight, or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts, and processes other than those specified in MIL-E-5400, it is desired their use be investigated. Where investigation shows advantages can be realized, a request for approval should be submitted to the procuring activity for consideration. Each request should be accompanied by complete supporting information.

6.4.1 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The First article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, and test approval of the documents' first article.

6.5 Maintenance module. A maintenance module is defined as the least complex segment of the contract end article(s) which is capable of being removed and replaced at organizational level maintenance using tools, support equipment, and procedures approved for the end articles(s). Further, it is the segment to which the Contractor should provide fault detection and isolation techniques consistent with scheduled and unscheduled organizational level maintenance requirements.

6.6 Nonrepairable subassemblies. As a general rule, nonrepairable subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high-voltage power supplies), the nonrepairable subassembly should evidence a Mean-Time-to-Failure greater than 5000 hours, and for many applications, this figure should be nearer 50,000 hours.

6.7 Preproduction equipment. Preproduction equipment should be defined as an equipment suitable for complete evaluation of form, fit, and performance. It is in final form in all respects, employs standard parts (or nonstandard parts approved by procuring activity), and is completely representative of final equipment.

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6.8 Associated equipment. This equipment is intended for use with the following equipments: AN/AIC-22(V), AN/ARR-72, AN/AQA-7, TD-900A/AS, AN/ARR-78 and AN/UYS-1.

Preparing Activity
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-85308(AS)	Recorder/Reproducer, Acoustic, AN/AQH-4(V)2

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional

8. DATE OF SUBMISSION *(YYMMDD)*

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